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# Blend fractions from the electronic free energy: a calibrated switching function for two-state interatomic potentials in tungsten

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## Abstract

Layered interatomic potentials capture electronic excitation at classical molecular-dynamics cost by blending two machine-learned potentials, fitted at two electronic smearing temperatures, through a per-atom auxiliary variable  $\lambda$ . The blend fraction is the load-bearing choice, and it has been made with a logistic in the local atomic temperature whose switching temperature and width are free parameters. We argue that  $\lambda$  is not free: it is the fraction of the electronic free-energy distortion of the potential-energy surface realised at the local electronic temperature, and we construct the switching function that follows. Three things come out of it. The transition width ceases to be a parameter — the Sommerfeld law makes the correct blend fraction leave the origin quadratically, so equilibrium fluctuations cannot toggle atoms without a width being chosen for that purpose. The calibration question separates from the electron–ion coupling question, which a single map from ionic temperature to blend fraction cannot do. And the scheme’s energy non-conservation, which we show is exactly  $d(K + U^{\text{gs}})/dt = \sum_j \lambda_j \Delta \mathbf{F}_j \cdot \mathbf{v}_j$ , becomes a design criterion rather than a caveat: a spatially uniform blend is exactly conservative, so the injected work is controlled by the spread of  $\lambda$  across the sample and therefore by the slope of the switching function. Calibrating against a finite-electronic-temperature electronic-structure calculation we perform for the purpose, the resulting switch for tungsten is  $11.2\times$  shallower than the published logistic at its steepest and, calibrated against that reference, reduces the energy error of the blended potential at an intermediate electronic temperature from  $1.62 \text{ meV atom}^{-1}$  to  $0.93 \text{ meV atom}^{-1}$ . That calibration fails a validation gate we declared in advance, so the cascades below are run with the citation-anchored alternative the pre-registration mandates instead — which against our own reference is  $3.5\times$  worse than ground-state-only dynamics, and against the published optimum is exact by construction. The two references disagree by a factor of 3–6 depending on the channel, and we could not adjudicate between them. In unthermostatted cascade dynamics it halves the total-energy excursion ( $0.0546$  against  $0.1047 \text{ eV atom}^{-1}$ ) and spreads  $\lambda$   $1.9\times$  less widely across the sample while carrying *greater* excited-state exposure; the irreducible non-conservative work falls by  $2.8\times$ , though at 3 repetitions that particular ratio is not resolved. Over 181 displacement cascades in tungsten it preserves the framework’s signature, reducing the surviving Frenkel-pair count by  $34 \pm 9\%$  relative to ground-state-only dynamics ( $z = -3.94$ ); the reproduced logistic reaches  $28 \pm 8\%$ , and paired run by run the two are not separable on damage. The one axis on which they behave differently at all is failure: at the top of the ladder 2 of the logistic’s 12 repetitions diverge numerically, with the cell’s energy moving by  $50.056 \text{ eV atom}^{-1}$ , where ours on the identical configurations move by  $0.122$  — though that difference is not significant ( $p = 0.48$ ), and extending this energy by half again did not make it so. Four results cut against the construction and we report them with the rest.

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The electronic free energy of tungsten is quadratic in electronic temperature over the range concerned, so the correct blend fraction at 0.1 eV comes out 3–6 times smaller than the previously reported optimum. Only about 3 % of the variance of the *force* difference between the two provided potentials is explained by the electronic-temperature distortion we compute, against 37 % of the energy difference — which, since the framework blends forces, deserves a direct test. The reduction in thermal-spike volume, the other half of the framework’s reported signature, is not resolved in our cascades for any switching function at any energy we could reach ( $-2 \pm 3$  %), and the measure turns out to be incapable of resolving it: on the same trajectories it returns a significant *increase* at a 5 Å alpha-shape probe radius and a significant *decrease* at 15 Å, so we draw no conclusion from it at all.

## 1 Introduction

A classical molecular-dynamics simulation moves ions on a single Born–Oppenheimer surface. That surface belongs to a particular electronic state, and in almost all such simulations it is the ground state. There are situations where this is simply the wrong surface. The one that motivates this paper is a displacement cascade in a plasma-facing metal: a 14.1 MeV fusion neutron hands tens of keV to a primary knock-on atom, which deposits it in a picosecond into a nanometre-scale volume. For the duration of the thermal spike the local electronic state is far from the ground state, and a ground-state potential has no way to know it.

The layered, or two-state Sommerfeld, construction of Wood, Koknat and Thompson Wood et al. [2026] addresses this at classical-molecular-dynamics cost. Two machine-learned potentials for the same element are fitted to first-principles data computed at two different electronic smearing temperatures — a ground state  $\mathcal{U}_{\text{gs}}$  and a high-temperature  $\mathcal{U}_{\text{ht}}$  — and a per-atom auxiliary variable  $\lambda_i$  blends their forces. In the radiation-damage application  $\lambda$  is driven by the local atomic temperature, and the reported effect is clear and physically sensible: sampling the high-temperature surface reduces both the peak thermal-spike volume and the number of surviving Frenkel pairs.

The construction leaves one thing undetermined, and it is the thing that decides everything. *What should  $\lambda$  be?* The original work answers with a logistic in the local atomic temperature,  $\lambda_i = 1/(1 + e^{-(T_i - T^*)/\sigma})$ , with a switching temperature  $T^*$  treated as a free parameter and a width fixed by fiat at  $\sigma = T^*/14$ ; it says so explicitly, notes that no sweep of  $\sigma$  was carried out, and reports that the resulting dynamics become unstable at high recoil energy for the more aggressive settings — the  $T^* = 0.5$  eV series has to be terminated at 15 keV while  $T^* = 5$  eV reaches 200 keV. It also reports a diagnostic that no re-tuning of  $T^*$  can fix: at an intermediate electronic temperature of 0.1 eV, the fixed blend fraction that minimises the energy error against density-functional theory is  $\lambda \approx 0.2$  while the one that minimises the force error is  $\lambda \approx 0.35$ .

**Thesis.**  $\lambda$  is not a switch to be tuned. It is a physical fraction — the share of the electronic free-energy distortion of the potential-energy surface that is realised at the local electronic temperature — and once it is fixed that way, three things follow that do not follow from a logistic.

First, **the transition width stops being a free parameter.** The leading electronic free energy of any metal is quadratic in the electronic temperature, so the correct blend fraction leaves the origin with vanishing slope. The property that the original work had to buy with  $\sigma = T^*/14$  — that equilibrium temperature fluctuations do not toggle atoms onto the excited surface — is a consequence of thermodynamics rather than a choice.

Second, **the accuracy question and the non-equilibrium question separate.** A logistic maps the local ionic temperature straight to a blend fraction, so one function has to do two jobs: say how excited the electrons are, and say what an excited electronic state does to the surface. Those are different physics, and they have different functional forms. Separating them lets the second be calibrated against electronic structure and the first be stated as an explicit, single-parameter model of electron–ion energy partition.

Third — and this is the part we did not anticipate — **the principled switch is also the gentler one, for a reason that can be written down exactly.** Because the blend is applied to forces rather than energies, the scheme does not conserve energy. We show in §2.2 that the departure from ground-state

dynamics is exactly  $d(K + U^{\text{gs}})/dt = \sum_j \lambda_j \Delta \mathbf{F}_j \cdot \mathbf{v}_j$ , and that a spatially *uniform* blend fraction is perfectly conservative — so the irreducible non-conservation is set by how much  $\lambda$  varies from atom to atom, and therefore by the *slope* of the switching function. A calibrated switch is broad and unsaturating where a logistic is narrow and saturating, so at matched excited-state exposure it varies less across a spike front and injects less energy — which we measure, and which is what a switch has to do if the instability is a price of switching too sharply rather than an unavoidable price of force blending. Whether that translates into a wider stable recoil range we could not establish: §4.5 reports two diverged repetitions of the reproduced logistic at the top of our ladder against none of ours on the same configurations, which is an existence proof and not a measured difference in rate.

**What this paper does.** We construct the free-energy-fraction (FEF) switch (§2.3), calibrate it against a finite-electronic-temperature electronic structure calculation for tungsten that we perform for this purpose (§3.2), and put it through the accuracy, stability and physical-effect tests that the layered framework demands (§4), against the published logistic reproduced under its own parameters. We also report four results that do not favour our own construction, because they are what the measurements say: the calibration we obtain from electronic structure disagrees with the reported intermediate-temperature optimum by a factor of 3–6 (§4.2); in the force channel our reference cannot resolve the distortion well enough to adjudicate at all (§4.3); the reduction in thermal-spike volume that the framework reports is not resolved in our cascades for any switching function, and that measure turns out to be unable to settle the question either way, returning a significant increase at one analysis probe radius and a significant decrease at another (§4.6); and the published criterion for terminating a recoil series never fires anywhere in the range we could afford, so what stability evidence we have rests on a failure mode that criterion does not detect (§4.5). The reason the comparison rests on a surrogate rather than on the first-principles reference sets is that those sets were not available to us; that is stated plainly in §3.2 and its consequences are traced through §8.

## 2 What a switching function has to do

### 2.1 The layered potential

Two interatomic potentials for the same element, fitted to first-principles data computed at two different electronic smearing temperatures, define a one-parameter family of potential-energy surfaces through a blend fraction  $\lambda$ ,

$$\mathcal{U}_{\text{tot}} = \lambda \mathcal{U}_{\text{ht}} + (1 - \lambda) \mathcal{U}_{\text{gs}}, \quad (1)$$

with  $\mathcal{U}_{\text{gs}}$  fitted at  $\theta_{\text{gs}} = 0.001$  eV and  $\mathcal{U}_{\text{ht}}$  at  $\theta_{\text{ht}} = 0.5$  eV (we write electronic temperatures as energies,  $\theta = k_{\text{B}} T_e$ ). In the layered implementation of Wood et al. [2026],  $\lambda$  is a per-atom variable and the blend is applied to the *forces*,

$$\mathbf{F}_j = - \sum_i \left[ \lambda_i \nabla_j \mathcal{U}_i^{\text{ht}} + (1 - \lambda_i) \nabla_j \mathcal{U}_i^{\text{gs}} \right], \quad (2)$$

where  $\mathcal{U}_i^{\text{ht}}$  and  $\mathcal{U}_i^{\text{gs}}$  are per-atom energies. Everything in this paper holds (1)–(2) and the two potentials fixed; the object of study is the rule that assigns  $\lambda_i$ .

Two demands are placed on that rule and they pull in opposite directions. The *accuracy* demand is that a region whose electrons are at temperature  $\theta$  should feel the potential-energy surface that belongs to  $\theta$ . The *stability* demand is that the dynamics generated by (2) not run away. The next two subsections make each demand quantitative; §2.3 then constructs a rule that meets both.

### 2.2 Exactly how much energy a force blend does not conserve

Equation (2) is not the gradient of any scalar, a point made but not quantified in the original work. It is worth being precise, because the answer turns out to be a design criterion rather than a caveat.

Write the two endpoint force fields as  $\mathbf{F}_j^{\text{gs}} = -\nabla_j U^{\text{gs}}$  and  $\mathbf{F}_j^{\text{ht}} = -\nabla_j U^{\text{ht}}$ , and their difference as  $\Delta \mathbf{F}_j = \mathbf{F}_j^{\text{ht}} - \mathbf{F}_j^{\text{gs}} = -\nabla_j \Delta U$  with  $\Delta U = U^{\text{ht}} - U^{\text{gs}}$ . What the per-atom implementation integrates is each atom’s *own* blend fraction multiplying its total sub-style force,

$$\mathbf{F}_j = \mathbf{F}_j^{\text{gs}} + \lambda_j \Delta \mathbf{F}_j. \quad (3)$$

Since  $\mathbf{F}_j^{\text{gs}}$  is a gradient,  $\dot{K} = \sum_j \mathbf{F}_j \cdot \mathbf{v}_j$  gives immediately

$$\boxed{\frac{d}{dt}(K + U^{\text{gs}}) = \sum_j \lambda_j \Delta \mathbf{F}_j \cdot \mathbf{v}_j = - \sum_j \lambda_j (\nabla_j \Delta U) \cdot \mathbf{v}_j.} \quad (4)$$

This is exact, requires no assumption about how  $\lambda$  is generated, and accounts for the *entire* departure of the scheme from the ground-state dynamics. Splitting  $\lambda_j = \bar{\lambda} + \delta\lambda_j$  into its instantaneous mean over the sample and the deviation from it,

$$\frac{d}{dt}(K + U^{\text{gs}} + \bar{\lambda} \Delta U) = \Delta U \frac{d\bar{\lambda}}{dt} - \underbrace{\sum_j \delta\lambda_j (\nabla_j \Delta U) \cdot \mathbf{v}_j}_{\text{irreducible}}. \quad (5)$$

Three things follow, and they shape the design.

**(i) Even at frozen blend fractions the scheme is non-conservative, and only because  $\lambda$  is non-uniform.** If  $\lambda_j$  were the same for every atom, the second term of (5) would vanish identically and the first would too at fixed  $\bar{\lambda}$ : a *global* blend is a perfectly good conservative potential, namely  $U^{\text{gs}} + \lambda \Delta U$ . Non-conservation is not the price of blending; it is the price of blending *differently in different places*. The irreducible term is proportional to the spatial spread of  $\lambda$  across the sample.

**(ii) The spurious power is linear in the slope of the switch.** With  $\lambda_j = \lambda(T_j)$ , the spread is  $\delta\lambda_j \simeq \lambda'(T)(T_j - \bar{T})$  to first order, so the irreducible term of (5) scales as  $\lambda'$  times the local-temperature contrast times the distortion force. The heating and cooling of a thermal spike, and  $\Delta \mathbf{F}$ , belong to the cascade and to the pair of potentials; the only thing the modeller controls is  $\lambda'$ . Every admissible switch satisfies  $\int_0^\infty \lambda'(T) dT = 1$  — they differ only in how that unit slope is distributed over the driver axis. A logistic with the published width rule  $\sigma = T^*/14$  concentrates it in a window of width  $\sim 4\sigma$ , with peak slope  $1/(4\sigma) = 3.5/T^*$ , and that window sits exactly where a spike front sweeps through. A switch that spreads the same unit slope over the whole sampled range has a peak slope smaller by the corresponding ratio.

**(iii) In the uniform limit, the injection is a hysteresis loop area.** It is worth recording the special case, because it is the intuitive picture. Define the frozen- $\lambda$  blended energy  $U_\lambda = \sum_i [\lambda_i \mathcal{U}_i^{\text{ht}} + (1 - \lambda_i) \mathcal{U}_i^{\text{gs}}]$  with  $\Delta U_i = \mathcal{U}_i^{\text{ht}} - \mathcal{U}_i^{\text{gs}}$ . When  $\lambda$  varies slowly in space,  $d(K + U_\lambda)/dt \simeq \sum_i \Delta U_i \dot{\lambda}_i$ , so an atom that heats and cools back receives

$$\delta E_i = \oint \Delta U_i d\lambda_i, \quad (6)$$

which vanishes only if  $\Delta U_i$  is the same on the two legs of the excursion. It is not: an atom in the molten core on the way up sits in a different local environment from the same atom in the recrystallised solid on the way down. The loop area is bounded by the traversed *range* of  $\lambda$  times the hysteresis in  $\Delta U$ . A switch that saturates drives every hot atom over the full range  $0 \rightarrow 1 \rightarrow 0$ ; one that does not saturate drives it over a fraction of that range.

**At matched excited-state exposure, longer and gentler beats brief and saturated.** The exposure diagnostic of the original work is  $\chi = \int \eta dt$  with  $\eta = \sum_i \lambda_i$ , so an atom reaching peak blend  $\lambda_i^{\text{max}}$  for a residency  $\tau_i$  contributes  $\chi_i \simeq \lambda_i^{\text{max}} \tau_i$  while contributing  $|\delta E_i| \lesssim \bar{\delta} \lambda_i^{\text{max}}$ , with  $\bar{\delta}$  the typical  $\Delta U$  hysteresis. Summing,

$$|\sum_i \delta E_i| \lesssim \bar{\delta} \frac{\chi}{\bar{\tau}}, \quad \bar{\tau} \equiv \frac{\sum_i \lambda_i^{\text{max}} \tau_i}{\sum_i \lambda_i^{\text{max}}}. \quad (7)$$

At matched  $\chi$ , the switch that delivers its exposure through many atoms held gently for a long time injects less than the switch that delivers the same exposure through fewer atoms driven briefly to saturation. This is the prediction §5 bears on — in the event it tests a stronger version of it, our switch carrying *more* exposure than the logistic rather than equal exposure — and it is why the accuracy-driven construction below turns out also to be the gentler one.

### 2.3 What sets the blend fraction

**The calibration curve.** At electronic temperature  $\theta$  the surface the ions move on is the Mermin free energy Mermin [1965]  $F(\mathbf{R}; \theta) = E(\mathbf{R}; \theta) - \theta S_{\text{el}}(\mathbf{R}; \theta)$ . Write its distortion away from the ground-state reference as  $D(\mathbf{R}; \theta) = F(\mathbf{R}; \theta) - F(\mathbf{R}; \theta_{\text{gs}})$ . The two-potential basis spans exactly one direction of distortion,  $\Delta = \mathcal{U}_{\text{ht}} - \mathcal{U}_{\text{gs}}$ , and the layered potential is *linear* in  $\lambda$ , so the best that any blend fraction can do is the projection

$$\lambda^*(\theta) = \frac{\langle D(\theta), \Delta \rangle}{\langle \Delta, \Delta \rangle}, \quad (8)$$

evaluated separately in the energy and the force channel over an ensemble of configurations. Equation (8) *defines* the calibration curve: it is the blend fraction a material at electronic temperature  $\theta$  should be given, and it is a measurable property of the two potentials and of the electronic structure, not a modelling choice. Because energies and forces are different linear functionals of the same  $D$ , there is no reason for the two channels to agree, and §4.2 shows they do not.

**Its functional form.** Two limits are fixed by thermodynamics. As  $\theta \rightarrow 0$  the Sommerfeld expansion gives  $D \rightarrow -\frac{\pi^2}{6}\theta^2 N(E_{\text{F}}; \mathbf{R})$  for any metal, so  $\lambda^* \propto \theta^2$ : *the calibration curve leaves the origin quadratically, with zero slope*. At  $\theta = \theta_{\text{ht}}$  the correct answer is  $\lambda = 1$  by construction. Departures from the  $\theta^2$  law at larger  $\theta$  measure structure in the electronic density of states within a few  $\theta$  of  $E_{\text{F}}$  — for tungsten, the pseudogap that makes the group-VI metals body-centred cubic and the antibonding peak about 2.6 eV above it. We therefore adopt the two-parameter family

$$\Lambda(\theta) = \frac{g(\theta) - g(\theta_{\text{gs}})}{g(\theta_{\text{ht}}) - g(\theta_{\text{gs}})}, \quad g(\theta) = \frac{\theta^2}{1 + (\theta/\theta_{\times})^{2-p}}, \quad (9)$$

which is  $C^\infty$  and monotone on  $\theta > 0$  and reproduces the two endpoints exactly. The shape function  $g$  leaves the origin as  $\theta^2$  — that is the Sommerfeld law, and it is what removes the width parameter — and turns over above  $\theta_{\times}$  towards an asymptotic  $\theta^p$ . The asymptote is a statement about the functional form, not about the range we work in: measured on the deployed parameters the local exponent of  $\Lambda$  falls from 2.1 just above  $\theta_{\text{gs}}$  to 0.4 at  $\theta_{\text{ht}}$ , so  $p$  sets where the crossover heads rather than a power the curve attains below the training endpoint. The shape parameters  $(\theta_{\times}, p)$  are fixed by least squares to the measured curve (8). That is the principle; it is not what we were able to deploy. The pre-registration made the fit conditional on a validation gate that our surrogate reference fails in the force channel, so the deployed pair is instead pinned to the one published blend fraction at the intermediate temperature, as the pre-registration directs on failure. §6.1 states the gate, the failure and the substitution; §4.2 reports the measured curve alongside, and the two disagree by a factor of 3–6.

**The driver.** In a displacement cascade there is no thermometer for the electrons, and the framework’s driver is the centre-of-mass-corrected local atomic temperature  $T_i$ . The electronic subsystem is heated out of equilibrium only by the local ionic *excess* above the ambient bath  $T_0$ , and only a fraction  $f$  of that excess is realised locally as electronic excitation — the rest is carried away by the very fast electronic conduction or never leaves the ions. With a  $C^\infty$  rectifier of width  $w = 200$  K — narrow against every temperature in a cascade, so it only smooths the corner at  $T_i = T_0$  —

$$\theta_e(T_i) = \theta_{\text{gs}} + f k_{\text{B}} \text{sp}(T_i - T_0), \quad \text{sp}(x) = w \ln(1 + e^{x/w}). \quad (10)$$

The single scalar  $f$  is a *slope*, not a threshold. It rescales the driver axis; it does not place a transition anywhere, and no value of it produces the abrupt turn-on that a switching temperature does.  $f \rightarrow 1$  is the local-equilibration limit  $T_e \rightarrow T_i$ ;  $f \rightarrow 0$  is the fast-conduction limit.

**The switch.** Composing, and saturating smoothly because the two-potential basis cannot represent  $\theta > \theta_{\text{ht}}$ ,

$$\lambda_i = S(\Lambda(\theta_e(T_i))), \quad S(x) = x(1 + x^n)^{-1/n}, \quad n = 32. \quad (11)$$

We call this the *free-energy-fraction* (FEF) switch. Its selection principle is a single sentence: *give every atom the blend fraction that the electronic free energy of the material demands at that atom’s local electronic temperature, and infer that temperature from the local ionic overheating through one partition slope*.

**Why it is automatically well behaved.** The properties that the original work had to impose by hand follow here from the construction. It is  $C^\infty$ , so derivative-continuous. It recovers  $\lambda \rightarrow 1$  as  $T_i \rightarrow \infty$ , and at the bath it is not exactly zero but negligible: the rectifier leaves  $\text{sp}(0) = w \ln 2$ , so  $\lambda(T_0) = 4.29 \times 10^{-4}$ , a factor of 816 below the blend fraction the intermediate electronic temperature calls for.  $\lambda \rightarrow 0$  holds exactly only as  $T_i \rightarrow 0$ , where it is  $2.3 \times 10^{-6}$ . It does not toggle atoms under equilibrium fluctuations — not because a width was chosen to be one fourteenth of a switching temperature, but because the electronic free energy is quadratic in the electronic temperature near the origin, so the switch turns on with vanishing slope. And by (5) its slope is spread over the whole sampled driver range rather than concentrated at a threshold, which by (7) is exactly the property that limits the spurious energy injection at fixed excited-state exposure.

### 3 Methods

#### 3.1 Potentials, driver and implementation

The two potentials are the author-provided linear atomic cluster expansion (ACE) models Drautz [2019], Lysogorskiy et al. [2021] for tungsten, fitted with FitSNAP Rohskopf et al. [2023] to VASP data at electronic smearing  $\theta_{\text{gs}} = 0.001$  eV and  $\theta_{\text{ht}} = 0.5$  eV, each with its Ziegler–Biersack–Littmark (ZBL) repulsive core Ziegler et al. [1985] at full strength. They are used exactly as provided: no refitting, no change of basis, no reweighting. We verified the provided files reproduce the published  $T = 0$  lattice constants (3.1759 Å for  $\mathcal{U}_{\text{gs}}$  and 3.1775 Å for  $\mathcal{U}_{\text{ht}}$ , against 3.176 Å and 3.178 Å reported). The two models share the ACE function set but not the radial basis (dispersion parameters 0.7075 and 0.5128), so the blend cannot be collapsed into a single coefficient interpolation and both models are evaluated every step.

The driver is the centre-of-mass-corrected local atomic temperature from LAMMPS compute ave/sphere/atom with a 5.28 Å cutoff, the value the original study states it used throughout, with comm\_modify cutoff 5.28. The same compute feeds every switching function so the head-to-head comparison is on an identical driver. (The example input shipped with the potentials uses 5.488 Å; we follow the paper text and note the discrepancy.)

The blend is applied to the forces through LAMMPS’ Plimpton [1995], Thompson et al. [2022] pair\_style hybrid/scaled with atom-style scale factors, the mechanism introduced in the original work; the ZBL core carries a constant scale of unity. The switching functions are LAMMPS variable ... atom expressions generated by code/cascade/make\_switch.py, which also contains the reference Python implementation used for every analytic check; the two agree to machine precision on the snapshot test in code/cascade/verify\_switch.py. We use the recursive ACE evaluator rather than the product evaluator (bit-identical energies and forces on our test, 1.57× faster).

#### 3.2 A finite-electronic-temperature reference

**The reference set is missing.** The accuracy axis of this study was designed around the author-provided VASP reference sets at  $T_e = 0.001, 0.1$  and 0.5 eV. *Those files were not available to us:* an exhaustive search of the working tree finds only the two ACE potentials, the EAM table and an example cascade input. No electronic-structure engine is available and no new first-principles calculation was possible. We recorded this before any run (Appendix A) and constructed the substitute described next. This is the single most important limitation of the present work and we return to it in §8.

**What we use instead.** We compute the finite-electronic-temperature surface directly, with an orthogonal Slater–Koster Slater and Koster [1954]  $d$ -band tight-binding model of body-centred-cubic tungsten: canonical  $dd\sigma : dd\pi : dd\delta = -6 : 4 : -1$  ratios, Harrison  $r^{-5}$  scaling Harrison [1980], a  $C^2$  taper to zero at 5.6 Å, and a Monkhorst–Pack Monkhorst and Pack [1976] mesh of the supercell Brillouin zone. At fixed electron number the Mermin free energy is  $F = \sum_{k\mathbf{n}} w_k g \varepsilon_{k\mathbf{n}} f_{k\mathbf{n}} - T_e S_{\text{el}}$  and the forces are Hellmann–Feynman,  $-\sum_k w_k g \text{Tr}[\rho(k) \partial H(k) / \partial \mathbf{R}_a]$ , the entropy and chemical-potential terms dropping out by the variational property of the Mermin functional.

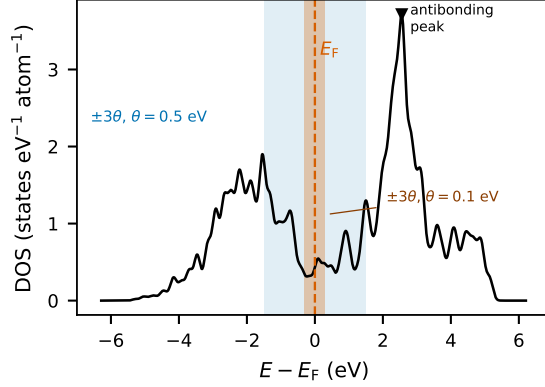


Figure 1: Electronic density of states of the calibrated tight-binding model for bcc tungsten. The Fermi level sits in the pseudogap that makes the group-VI transition metals body-centred cubic, at a  $d$ -band filling of 4.09 electrons and a density of states of  $0.436 \text{ states eV}^{-1} \text{ atom}^{-1}$ ; the shaded bands are  $\pm 3\theta$  at  $\theta = 0.1$  and  $0.5 \text{ eV}$ . The intermediate electronic temperature probes only the flat pseudogap; the training endpoint reaches  $1.5 \text{ eV}$ , onto the rising flank below the antibonding peak at  $2.6 \text{ eV}$ .

Two features make this defensible despite the crudeness of the model. First, *we never use an absolute energy*: every quantity is a difference between two electronic temperatures at fixed ionic positions, and in a non-self-consistent tight-binding total-energy model the pair-repulsive and double-counting terms do not depend on  $T_e$ , so they cancel identically in that difference. The model has to get the *change* of the surface with electronic smearing right, not the cohesive energy. Second, its two free scalars are fixed by measured properties of tungsten rather than tuned: the hopping scale by the  $d$ -band width ( $10.5 \text{ eV}$ ) and the filling by the band density of states at the Fermi level ( $0.430 \text{ states eV}^{-1} \text{ atom}^{-1}$ , from the measured electronic heat-capacity coefficient corrected for electron-phonon enhancement Lin et al. [2008]). The resulting model places  $E_F$  in the bcc pseudogap at a  $d$ -band filling of 4.13 electrons, with the antibonding peak about  $2.6 \text{ eV}$  above it, as tungsten does Lin et al. [2008]. Analytic forces agree with finite differences of the free energy to six significant figures (code/derivation/tb\_finite\_te.py).

Figure 1 is the resulting density of states, and it contains the physics that the rest of the paper turns on. Smearing at  $\theta = 0.1 \text{ eV}$  redistributes occupation only within a few tenths of an electronvolt of  $E_F$ , where the density of states is flat and low — the Sommerfeld regime, in which the free energy falls as  $\theta^2$ . Smearing at  $\theta = \theta_{\text{ht}}$  reaches the rising flank below the antibonding peak, about  $2.5 \text{ eV}$  above  $E_F$ , and the free energy then falls faster than  $\theta^2$ . A calibration curve that interpolates the two therefore cannot be linear in  $T_e$ , and §4.2 shows it is not.

We report a sensitivity sweep over both calibration targets in §4.2, and we state a validation gate, declared before the calculation: the reference is used only if its distortion at  $\theta_{\text{ht}}$  correlates with the ACE inter-potential distortion.

### 3.3 The blend-fraction sweep

The evaluation ensemble is 94 tungsten configurations of 53–55 atoms generated by  $\mathcal{U}_{\text{gs}}$  molecular dynamics and independent of any cascade: thermal bcc at 300, 1000, 2000 and 3000 K, over-heated and molten states at 4000 and 6000 K, isotropically strained cells at  $\pm 1.5\%$  and  $\pm 3\%$ , a monovacancy and a  $\langle 111 \rangle$  split interstitial each at 1000 and 2500 K, and rattled cells. For each configuration and each of  $T_e \in \{0.001, 0.01, 0.02, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5 \text{ eV}\}$  we sweep the global blend fraction over the 21 declared values  $\lambda \in \{0, 0.05, \dots, 1\}$  and record the root-mean-square error of the blended prediction against the reference distortion, in the energy channel (eV/atom, offsets removed) and the force channel (eV/Å, every component of every atom). Both channels are always reported. Because the blend is exactly linear in  $\lambda$  each sweep is a parabola, and we also report its closed-form minimiser (8).

### 3.4 Cascade protocol

Single-crystal bcc tungsten, cubic cells, equilibrated at 1000 K and 1 atm by Nosé–Hoover thermostat and barostat before the recoil; eight starting configurations 1 ps apart are emitted per cell and shared by every method. The primary knock-on atom is the atom nearest the cell centre (offset  $< 0.2 \text{ \AA}$ ); recoil directions are drawn isotropically from one fixed list, so repetition  $k$  of every method uses the same starting configuration *and* the same direction — the comparison is paired. Atoms within 2.5% of the cell edge from any face are held at 1000 K by a Nosé–Hoover thermostat and the interior is microcanonical, as in the original protocol. The ballistic and thermal-spike phase uses `fix dt/reset 100 1.0e-9 5.0e-4 0.05`, i.e. adaptive time steps from  $10^{-6}$  fs to 0.5 fs updated every 100 steps; the phase ends when the time step recovers its ceiling. The whole cell is then held at 1000 K for the anneal.

Surviving Frenkel pairs are counted by OVITO Stukowski [2010] Wigner–Seitz analysis against the ideal lattice of the same cell. The thermal-spike volume is the enclosed solid volume of an alpha-shape surface mesh Stukowski [2014] constructed over the atoms that polyhedral template matching Larsen et al. [2016] does not classify as bcc, with the probe sphere radius  $R_{\text{probe}} = 1 \text{ nm}$  used in the original study; §4.6 reports the sensitivity to that radius. We also record  $\eta(t) = \sum_i \lambda_i$  every ten steps and integrate it to  $\chi = \int \eta dt$  in atoms·ps, both to the end of adaptive stepping and to the end of the anneal.

**Declared deviations from the original protocol.** Three, all forced by compute and all applied identically to every method. (i) *Cell size and recoil range.* The original used cells of  $2.5 \times 10^5$  to  $1.6 \times 10^7$  atoms and recoil energies to 200 keV on leadership-class hardware. One 32-vCPU node evaluating two ACE potentials sustains  $2.2 \times 10^5$  atom step  $\text{s}^{-1}$  (measured), so we use cells of 5488–43904 atoms and the recoil ladder given in Table 2. Absolute defect counts are therefore not comparable to the original and we never present them as such; every comparison is between methods in the same cell at the same energy. (ii) *Anneal.* 8 ps at a 1 fs time step instead of 30 ps at 0.5 fs. We report the Frenkel-pair count as a function of anneal time so that how far recombination has proceeded is visible rather than assumed. (iii) *Repetitions.* 8 random orientations per (method, energy) instead of 25, extended to 12 at the top of the ladder for the two switching functions and the ground state (13 for the ground state, the extra one having no partner and entering no paired statistic) in an attempt to resolve the stability comparison of §4.5. The number completed at each energy is given in Table 2. Every completed repetition is reported; the two that diverged numerically are excluded from the statistics, with the exclusion and its effect stated in that table’s caption.

### 3.5 Baselines

Every comparison carries two references. The *as-reported logistic* is the original work’s own results, quoted with the figure or table they come from. The *reproduced logistic* is our own runs of  $\lambda_i = 1/(1 + e^{-(T_i - T^*)/\sigma})$  with the published width rule  $\sigma = T^*/14$  at the original’s own  $T^*$  values. Since the driver `c_lavg[2]` is in kelvin while the original quotes  $T^*$  in electronvolts without giving the conversion, we use  $T^*/k_B$ : 0.50 eV  $\rightarrow$  5802 K ( $\sigma = 414 \text{ K}$ ) and 5.00 eV  $\rightarrow$  58,022 K ( $\sigma = 4144 \text{ K}$ ), the extreme pair of the original’s set  $\{0.50, 0.75, 1.25, 2.50, 3.75, 5.00 \text{ eV}\}$ . Both baselines use the original’s own width rule and its own switching temperatures. On the accuracy axis we additionally report the *best possible* logistic under the published width rule, with  $T^*$  re-optimised against the same objective, and the best logistic with  $\sigma$  freed as well — upper bounds on what the logistic *form* can achieve. Ground-state-only dynamics ( $\lambda \equiv 0$ ) is the physical-effect reference and the Marinica embedded-atom-method (EAM) Marinica et al. [2013] tungsten potential is the classical comparison.

## 4 Results

### 4.1 The switch, and that it behaves

Figure 2 shows the deployed switch against the logistic at three of the original’s  $T^*$  values, on the driver they share. The deployed parameters are  $\theta_x = 0.100 \text{ eV}$ ,  $p = 0.2581$  (calibration, §4.2) and  $f = 0.14$  (partition slope, §4.4).

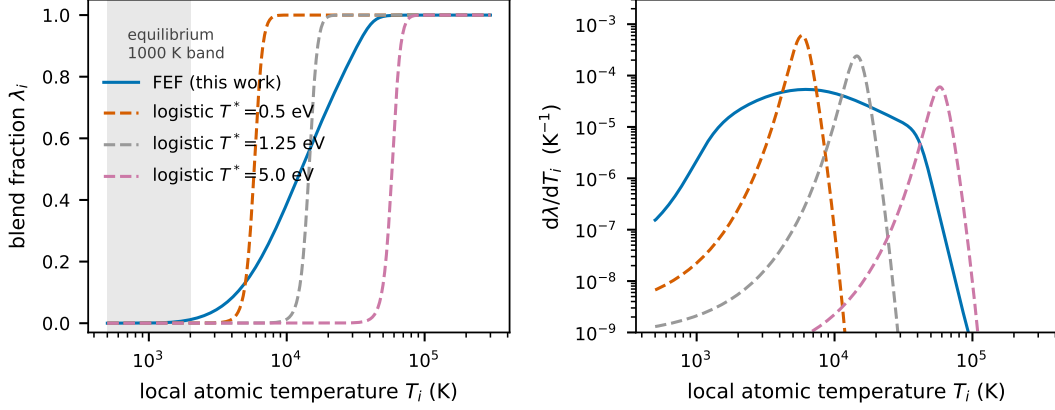


Figure 2: The deployed free-energy-fraction switch and the published logistic at three  $T^*$  values, as functions of the local atomic temperature they both consume. **Left:** blend fraction; the shaded band is a fixed 500 K to 2000 K bracket drawn for orientation, comfortably wider than the measured equilibrium local-temperature distribution at 1000 K (mean 972 K, standard deviation 129 K, largest of 336 000 samples 1689 K). **Right:** the slope, which §2.2 identifies as the quantity controlling the spurious power. Both axes are logarithmic; the logistic concentrates its entire unit slope integral in a window of width  $4\sigma = T^*/3.5$ .

Everything the framework asks of a switching function holds, and holds analytically rather than by construction of a width. The function is monotone on the whole driver range (checked on a 1 K grid to 200,000 K); its one-sided difference quotients agree to  $2.1 \times 10^{-11} \text{ K}^{-1}$ , so it is  $C^1$  to numerical precision; it tends to 0 as  $T_i \rightarrow 0$  ( $\lambda = 2.3 \times 10^{-6}$ ) and to 1 in the hot limit. At the 1000 K bath it is 0.0004, and at 1600 K — beyond the far tail of the equilibrium local-temperature distribution — it is 0.0048. Measured on 336 000 local-temperature samples from 20 ps of 1000 K equilibrium dynamics in a 16,000-atom cell (mean  $T_i = 972 \text{ K}$ , standard deviation 129 K, full width at half maximum 303 K, largest single sample 1689 K), the mean blend fraction is  $4.07 \times 10^{-4}$  and the largest value any atom ever reaches is  $5.26 \times 10^{-3}$ : not one of the 336 000 samples exceeds 0.01, and the expected instantaneous excitation of the whole cell is 6.51 atoms. The logistic at  $T^* = 0.5 \text{ eV}$  reaches  $4.90 \times 10^{-5}$  on the same field. Neither toggles; the difference is that the logistic achieves it because  $\sigma = T^*/14$  was chosen to be about the width of this distribution — and 303 K against  $\sigma = 414 \text{ K}$  confirms the original’s reasoning — whereas the free-energy-fraction switch achieves it because the electronic free energy is quadratic in the electronic temperature near the origin, with no width to choose.

The right panel of Fig. 2 is the quantity that will matter in §5. The logistic with the published width rule has maximum slope  $1/(4\sigma) = 3.5/T^*$ :  $6.03 \times 10^{-4} \text{ K}^{-1}$  at  $T^* = 0.5 \text{ eV}$ , concentrated in a window of width  $4\sigma$  around  $T^*$ . The deployed switch has maximum slope  $5.38 \times 10^{-5} \text{ K}^{-1}$ , at 6224 K, and spreads the same unit integral over the whole sampled range:  $11.2\times$  shallower at its steepest.

Two checks confirm that the blended surface itself is well behaved and that our pipeline reproduces the original’s own numbers. The W–W dimer binding curve under a global blend fraction interpolates between well depths of  $-3.145 \text{ eV atom}^{-1}$  at  $\lambda = 0$  and  $-1.826 \text{ eV atom}^{-1}$  at  $\lambda = 1$  — our own evaluation of the provided potentials, the original plotting the corresponding curves without tabulating the depths — and does so *exactly* linearly, to  $7.1 \times 10^{-15} \text{ eV}$ , as it must because the blend is linear at fixed geometry. Formation energies of metastable phases, which are computed at the relaxed geometry for each  $\lambda$  and so need not be linear, vary smoothly and monotonically: the fcc–bcc formation energy runs from  $0.491 \text{ eV atom}^{-1}$  to  $0.568 \text{ eV atom}^{-1}$ , against 0.491 and 0.568 in the original’s own table, departing from a straight line by at most  $0.004 \text{ eV atom}^{-1}$ ; the simple-cubic value runs from  $1.416$  to  $1.683 \text{ eV atom}^{-1}$  with a curvature of 0.027.

The LAMMPS realisation was checked against the Python reference implementation atom by atom on a configuration in which a recoil had been started, so the driver spans 1000 K to  $3.54 \times 10^5 \text{ K}$  across 5488 atoms: the largest discrepancy in  $\lambda_i$  is  $5.0 \times 10^{-13}$  and the largest violation of  $\lambda_i + (1 - \lambda_i) = 1$  as the two scale factors actually handed to `pair_style hybrid/scaled` is 0.

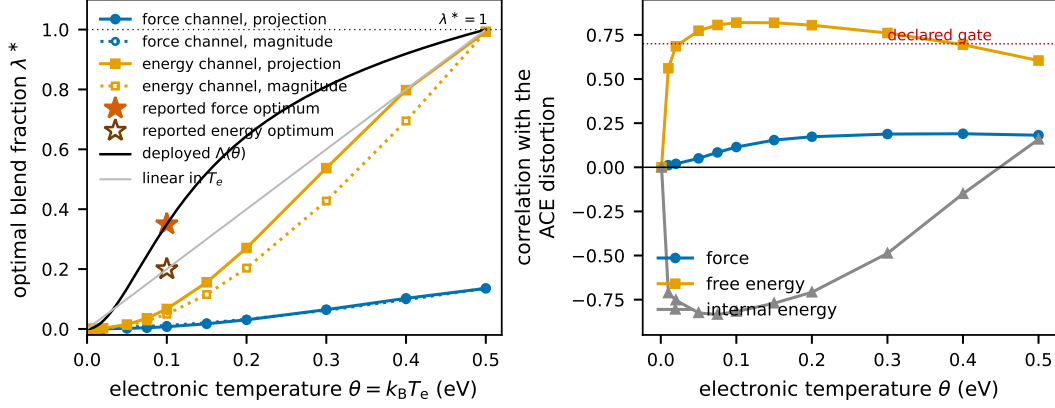


Figure 3: **Left:** the optimal blend fraction against electronic temperature, measured on 94 tungsten configurations with the finite-electronic-temperature reference, in both channels and with both estimators, together with the deployed calibration curve, the two optima reported by Wood *et al.* at 0.1 eV, and the naive linear-in- $T_e$  line. The curves are absolute, not normalised to the endpoint, so the endpoint check is legible directly: at  $\theta = \theta_{ht}$  the answer must be 1, and the energy channel reaches 0.993 while the force channel reaches only 0.136. That gap is the failure §6.1 reports, and it is why the deployed calibration is anchored rather than fitted. **Right:** correlation between the reference distortion and the ACE inter-potential distortion, with the declared validation gate. The internal-energy channel anti-correlates, which is how we know the training energies behave like free energies rather than bare internal energies.

#### 4.2 Calibration: what the blend fraction should be

The two provided potentials differ by  $\Delta_E = \mathcal{U}_{ht} - \mathcal{U}_{gs}$ , whose configuration-dependent part has a root-mean-square of  $19.8 \text{ meV atom}^{-1}$  across the ensemble (on top of a constant offset of  $568 \text{ meV atom}^{-1}$  that reflects the two fits' different energy references and is removed throughout), and by  $\Delta_F$  with a root-mean-square of  $0.306 \text{ eV \AA}^{-1}$  per force component, against a typical force magnitude of about  $1.7 \text{ eV \AA}^{-1}$ . The blend can move the prediction along that one direction and no other, so the whole calibration question is: how far along it should a material at electronic temperature  $\theta$  be moved?

Figure 3 answers it with the tight-binding reference. *The endpoint check passes in the energy channel and fails in the force channel.* At  $\theta = \theta_{ht}$ , where the answer must be 1, the measured projection is 0.993 in the energy channel — an independent electronic-structure calculation, told nothing about the two potentials, reproducing the endpoint to 1 %. In the force channel it is 0.136 with a correlation of 0.182: the reference force distortion has a comparable magnitude but points, per atom, largely elsewhere. The declared gate ( $r \geq 0.7$ , evaluated at the endpoint) is failed outright by the force channel and, by 0.095, by the energy channel. The right panel of Fig. 3 shows why that last statement needs care: the energy-channel correlation is above the gate over most of the grid and peaks at 0.820 at the intermediate temperature itself; it is only at the  $\theta_{ht}$  endpoint, where the gate happens to be evaluated, that it falls to 0.605. We nonetheless treat the gate as failed, because that is what we declared (§6).

Resolving the correlation by regime is informative: in near-equilibrium configurations the force-channel correlation is 0.368 and the implied endpoint blend fraction 0.344, and both collapse in the molten configurations ( $r = 0.015$ ). The energy channel behaves in the opposite direction. Read one way, the  $d$ -band surrogate simply cannot resolve the per-atom force field of a disordered hot metal; read the other, most of the force difference between the two potentials is not electronic-temperature physics. We cannot separate the two without the reference sets.

*The shape of the calibration curve is robust and it is not linear in  $T_e$ .* At the intermediate temperature the measured optimum is 0.0547 (force channel) and 0.0673 (energy channel), and the direction-independent magnitude estimator gives 0.0866 and 0.0496. All four are near the Sommerfeld value  $(T_e/\theta_{ht})^2 = 0.04$  and far from the linear value 0.2. The declared sensitivity sweep — 7 variants of the surrogate spanning the  $d$ -band width from 9.5 to 11.5 eV,  $N(E_F)$  from 0.35 to 0.55 states  $\text{eV}^{-1} \text{ atom}^{-1}$ , and  $k$ -meshes from  $3^3$  to  $5^3$  — moves the force-channel value only over

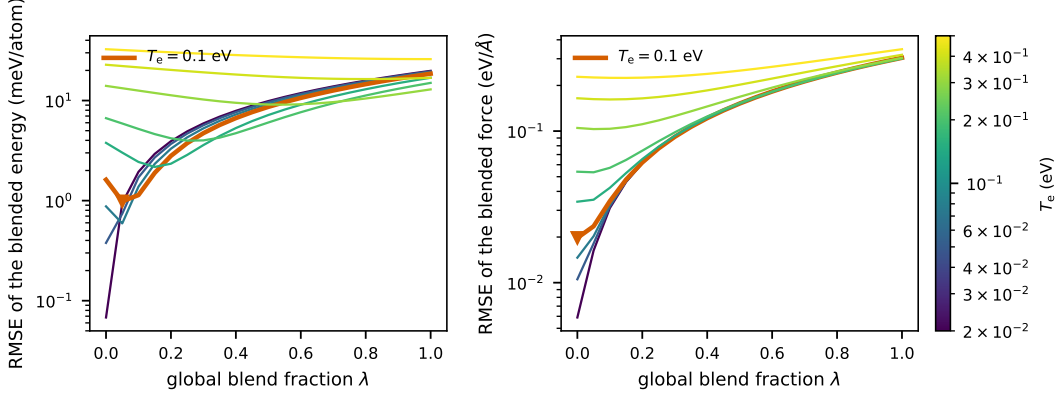


Figure 4: Blend-fraction sweep. Root-mean-square error of the blended prediction against the finite-electronic-temperature reference over the 21 declared blend fractions, for every electronic temperature on the grid above 0.02 eV (the two coldest are omitted; their curves are flat on this scale); the held-out intermediate temperature is highlighted. Left: energy channel. Right: force channel. The squared error is exactly quadratic in  $\lambda$  because the blend is exactly linear in it; the axis here is the root-mean-square error on a log scale.

0.030–0.060 and the energy-channel value over 0.016–0.084. The largest force-channel value anywhere in the sweep is still  $6\times$  below the reported force optimum, and the largest value in either channel is  $4\times$  below it, so the gap is not a calibration artefact. The sweep also provides an independent check on the targets we chose: the endpoint blend fraction, which must be unity, ranges over 0.73–1.03 across the variants, and the variant closest to unity is the one whose two targets are set to the measured properties of tungsten rather than perturbed away from them. This is what the thermodynamics requires: the leading electronic free energy of a metal is quadratic in the electronic temperature, and tungsten at 0.1 eV is still close to that regime. It is also a factor of four below the optima reported for the same intermediate temperature. §6 treats the disagreement at length rather than choosing a winner.

*Energy and force optima differ, but not in the reported direction.* Figure 4 is the blend-fraction sweep, the direct analogue of the original’s own study. At  $T_e = 0.1$  eV the energy-channel error is minimised on the declared grid at  $\lambda = 0.05$ , falling from  $1.617$  meV atom $^{-1}$  at  $\lambda = 0$  to  $0.9843$  meV atom $^{-1}$ , while the force-channel error is minimised at  $\lambda = 0.00$  (closed-form minimiser 0.0074 — the unnormalised projection, to be read against the endpoint-normalised 0.0547 quoted above — against 0.0668 for the energy channel). So the two channels do disagree, and no single fixed blend fraction is simultaneously energy- and force-optimal — which is the phenomenon the original reports. The *ordering* is the other way round: against this reference the force channel wants *less* blending than the energy channel, where the reported optima have the force channel wanting more. We take this as further evidence that the force channel of our surrogate is not resolving the same object as the reported force optimum, and not as a refutation of it.

### 4.3 Accuracy at the intermediate electronic temperature

Table 1 and Fig. 5 put every scheme on the same axis. The comparison is deliberately generous to the baseline: besides the original’s own  $T^*$  values we report the best logistic under its own width rule with  $T^*$  re-optimised against the same objective, and the best logistic with  $\sigma$  freed as well.

In the energy channel the result is unambiguous. Every logistic under the published width rule assigns  $\lambda < 10^{-4}$  at 0.1 eV (freeing  $\sigma$  raises it only to 0.0174), because a function that must be negligible at the 1000 K bath and saturated in a thermal spike has nothing left with which to be of order  $10^{-2}$  at 1160 K, the ionic-temperature equivalent of the held-out electronic temperature. Freeing  $\sigma$  raises the best logistic only to  $\lambda = 0.0174$ . The free-energy-fraction switch *under our own measured calibration* assigns 0.0627 and reduces the energy error at the intermediate temperature from  $1.62$  meV atom $^{-1}$  to  $0.93$  meV atom $^{-1}$ , and the integrated error from 11.6 to 8.67, which is within 0.5% of the per-temperature oracle.

Table 1: Error of the blended potential at the held-out intermediate electronic temperature  $T_e = 0.1$  eV, and integrated over the electronic-temperature grid, against the finite-electronic-temperature reference. Every scheme is asked the same question: what blend fraction does it assign to a material at electronic temperature  $T_e$ ? Lower is better; the last row is the unattainable per-temperature optimum. The anchored row is the switch the cascades of §4.5 were actually run with; against this reference it is the worst row in the table, which §4.3 and §6.1 discuss.

|                                          | $\lambda(0.1 \text{ eV})$ | at 0.1 eV            |                 | integrated           |                 |
|------------------------------------------|---------------------------|----------------------|-----------------|----------------------|-----------------|
|                                          |                           | energy<br>(meV/atom) | force<br>(eV/Å) | energy<br>(meV/atom) | force<br>(eV/Å) |
| FEF, measured calibration                | 0.0627                    | 0.93                 | 0.0259          | 8.67                 | 0.1385          |
| FEF, anchored ( <i>deployed</i> )        | 0.3500                    | 5.7                  | 0.107           | 10.5                 | 0.1934          |
| logistic, $T^* = 0.5 \text{ eV}$         | 0.0000                    | 1.62                 | 0.0197          | 11.6                 | 0.09321         |
| logistic, $T^* = 5.0 \text{ eV}$         | 0.0000                    | 1.62                 | 0.0197          | 12.2                 | 0.09149         |
| best logistic, $\sigma = T^*/14$         | 0.0000                    | 1.62                 | 0.0197          | 11.7                 | 0.09172         |
| best logistic, $\sigma$ free             | 0.0174                    | 1.35                 | 0.0199          | 10.8                 | 0.09241         |
| ground state only ( $\lambda \equiv 0$ ) | 0.0000                    | 1.62                 | 0.0197          | 12.2                 | 0.09149         |
| per- $T_e$ optimum (oracle)              | 0.0668                    | 0.926                | 0.0196          | 8.63                 | 0.08996         |

*That is not the switch the cascades were run with, and the distinction is not cosmetic.* Because the validation gate failed (§6.1), the deployed calibration is the citation-anchored one,  $\lambda(0.1 \text{ eV}) = 0.3500$ , and against *this* reference it is the worst row in Table 1 at the held-out temperature itself, in both channels:  $5.7 \text{ meV atom}^{-1}$  in the energy channel against 1.62 for ground-state-only dynamics, a factor of 3.5 worse rather than better. The two calibrations differ because they are answering to two different references, and the size of that difference is the paper’s central negative result rather than a detail of it: our surrogate says the correct blend fraction at 0.1 eV is 0.0627, the published optimum says 0.3500, and no switching function can be simultaneously right for both. We deployed the anchored one because the pre-registration says to defer to the published anchor when our own reference fails its gate, and we report the accuracy of both.

In the force channel it is not. Against this reference the per-temperature oracle is only 1.7% better than  $\lambda \equiv 0$ , because the reference force distortion is poorly aligned with the direction the blend can move in; the switch that tracks the measured optimum therefore cannot show a force-channel gain, and does not. We report this rather than suppressing the channel. Against the anchored calibration — that is, if the reported intermediate-temperature optimum is taken as the target — the ordering is reversed: the anchor is pinned to the reported *force* optimum,  $\lambda(0.1 \text{ eV}) = 0.3500$ , so the anchored switch is exact in that channel by construction and 75 % high against the reported energy optimum of 0.20, while every logistic sits at  $\lambda \approx 0$  and incurs the full  $|\lambda^*| \|\Delta\|$  error in both. That no single blend fraction can satisfy both reported optima at once is the original’s own observation and the reason this paper exists. Which of these two statements one believes is exactly the question §6 cannot settle.

Two structural remarks do not depend on which reference is used. First, the required calibration is nonzero and rising already at ordinary temperatures, while a switch deployed in a cascade must give  $\lambda \approx 0$  for an atom sitting at the bath. A single map from ionic temperature to blend fraction cannot do both; that is not a matter of choosing  $T^*$  better. The free-energy-fraction switch separates the two maps, and the driver map sends the bath to  $\theta_{\text{gs}}$  by construction. Second, no logistic can reproduce the *shape* of the calibration: it is a sigmoid in  $T_e$  where the required curve is a power law leaving the origin quadratically, and the best free- $\sigma$  logistic ( $T^* = 0.542 \text{ eV}$ ,  $\sigma = 1270 \text{ K}$ ) buys its intermediate-temperature accuracy by assigning a finite blend fraction at zero temperature.

#### 4.4 Setting the partition slope, and matched aggressiveness

The one remaining scalar is the partition slope  $f$ . A local two-temperature balance,  $G(T_i - T_e) \simeq (\kappa_e/L^2)(T_e - T_0)$ , gives  $f = GL^2/(\kappa_e + GL^2)$ ; with tungsten’s electronic thermal conductivity  $\kappa_e \simeq 125 \text{ W m}^{-1} \text{ K}^{-1}$  at 800 K Zarkadoula et al. [2015] and an electron–phonon coupling between the ab-initio  $G \simeq 1.7 \times 10^{17} \text{ W m}^{-3} \text{ K}^{-1}$  Daraszewicz et al. [2014], Lin et al. [2008] and the value the cascade literature uses,  $7 \times 10^{17}$  Fujimoto et al. [1984], a spike of radius 2 to 10 nm gives  $f$  between about 0.005 and 0.36. That bracket is too wide to fix anything, mostly because  $G$  for

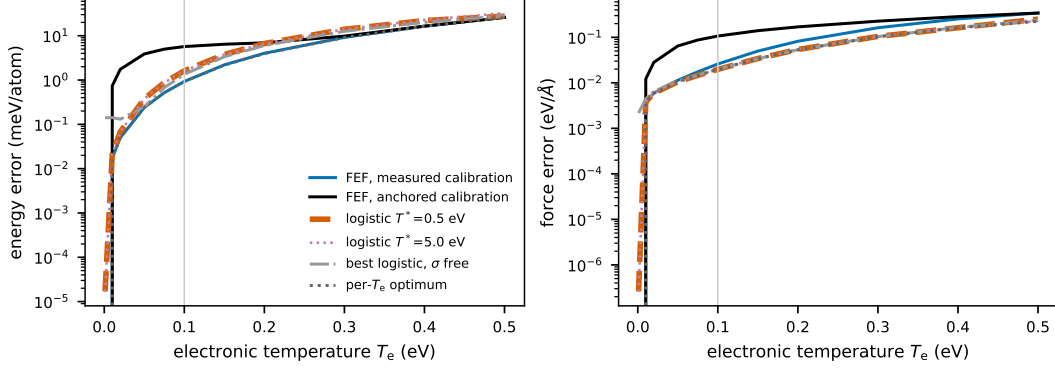


Figure 5: Error of the blended potential against the finite-electronic-temperature reference as a function of electronic temperature, in the energy (left) and force (right) channels, for each switching function. The vertical line is the held-out intermediate temperature.

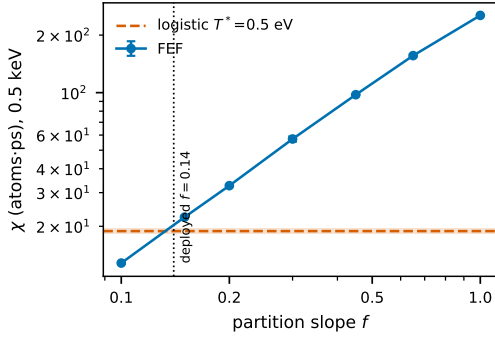


Figure 6: Setting the partition slope. Excited-state exposure  $\chi = \int \sum_i \lambda_i dt$  at 0.5 keV for each value of the partition slope  $f$  (every declared point, plotted as the mean of 2 repetitions with its standard error), against the reproduced logistic at  $T^* = 0.5$  eV (dashed, its standard error shaded and narrower than the line). The curves cross at  $f = 0.133$ ; we deployed the round  $f = 0.14$  just above it, which overshoots the logistic’s exposure by 8 %. These scan runs use a 1 ps anneal against the 8 ps of the production cascades, so the  $\chi$  here is not the  $\chi$  of Table 2; §4.5 discusses what that costs.

tungsten is itself uncertain by a factor of four, and it excludes the non-thermal electronic excitation that the layered construction exists to capture.

We therefore fix  $f$  the way the comparison requires: by matching the excited-state exposure  $\chi$  of the reproduced logistic at  $T^* = 0.5$  eV, at the lowest energy of the ladder, and then holding that single value for every energy and every other experiment. Figure 6 is the declared scan, 7 values of  $f$  from 0.10 to 1.00 with 2 repetitions each; every point is shown. The logistic’s exposure is  $\chi = 18.88 \pm 0.58$  atoms·ps, the log-log interpolated crossing is  $f = 0.133$ , and we deployed the round value  $f = 0.14$  just above it, which overshoots the logistic’s exposure by 8 %. It falls inside the physical bracket, at its upper end.

#### 4.5 Stability across the recoil ladder

Table 2 is the full cascade result and Fig. 7 the stability picture. The comparison is at matched aggressiveness by construction, but the match depends on where the exposure integral is truncated and the two switches distribute their exposure differently in time. The logistic’s  $\eta$  collapses as soon as the spike cools below  $T^*$ , so its  $\chi$  saturates almost immediately; the free-energy-fraction switch has a long low- $\lambda$  tail that keeps accumulating. At 0.5 keV, integrated to the end of the anneal, the free-energy-fraction switch delivers  $\chi = 38.76$  atoms·ps against 21.28 for the logistic; integrated to the end of adaptive time stepping, the same runs give 10.91 against 18.35. The switch that is more aggressive on one convention is less aggressive on the other. Part of that is our own doing and we state it: the partition-slope scan that fixes  $f$  (§4.4) was run with a 1 ps anneal against the 8 ps of the production cascades, so  $f$  was matched on an integral truncated far earlier than the one Table 2 reports. Re-running the scan at the production length would have cost another campaign and we did not do it. Rather than choose a convention after the fact we ran a second variant at  $f = 0.25$ , which matches the logistic on the second convention (18.60 against 18.35), and report both. On the adaptive-stepping convention the two variants bracket the logistic (10.91 and 18.60 against 18.35);

Table 2: Displacement cascades in bcc tungsten. Mean with the standard error of the mean over the converged repetitions, with the last column giving the number that failed out of the number run; every repetition of every method uses the same starting configuration and the same recoil direction, so the differences are paired. The ground-state and EAM references carry  $\lambda \equiv 0$  by construction, so their measured  $\chi$  and  $\eta_{\text{peak}}$  are exactly zero.  $\chi$  is integrated to the end of the anneal; the value integrated to the end of adaptive time stepping is in `logs/cascade_summary.json` and is discussed in §4.5.  $\chi$  is the excited-state exposure  $\int \sum_i \lambda_i dt$  in atoms-ps; “flag” is the number of repetitions that failed, under either the declared spike-volume criterion or the total-energy-divergence detector of §4.5; a series terminates when that exceeds a quarter of them. *Every* column is taken over the converged repetitions only, exposure included: a diverged trajectory is unphysical for its whole length, so the  $\chi$  it accumulates measures nothing. This matters in one cell and we flag it rather than leave it to be discovered — over all 12 repetitions the 5 keV logistic reads  $\chi = 667.11$  and  $\eta_{\text{peak}} = 1774$ , above our switch rather than below it, so the exclusion is what puts our switch at the higher exposure there.

| $E_{\text{PKA}}$ (cell) | method                                   | Frenkel pairs   | $V_{\text{peak}}$ (nm <sup>3</sup> ) | $\chi$ | $\eta_{\text{peak}}$ | flag |
|-------------------------|------------------------------------------|-----------------|--------------------------------------|--------|----------------------|------|
| 0.5 keV<br>5 488 atoms  | FEF, $f = 0.14$                          | $1.12 \pm 0.23$ | $3.55 \pm 0.20$                      | 38.76  | 98                   | 0/8  |
|                         | FEF, $f = 0.25$                          | $1.38 \pm 0.18$ | $3.55 \pm 0.20$                      | 90.17  | 158                  | 0/8  |
|                         | logistic $T^* = 0.5$ eV                  | $1.38 \pm 0.18$ | $3.60 \pm 0.22$                      | 21.28  | 193                  | 0/8  |
|                         | logistic $T^* = 5.0$ eV                  | $0.88 \pm 0.12$ | $3.28 \pm 0.14$                      | 1.19   | 41                   | 0/8  |
|                         | ground state only ( $\lambda \equiv 0$ ) | $1.38 \pm 0.37$ | $3.56 \pm 0.16$                      | 0.00   | 0                    | 0/8  |
|                         | EAM (Marinica)                           | $1.75 \pm 0.16$ | $5.38 \pm 0.51$                      | 0.00   | 0                    | 0/8  |
| 1 keV<br>11 664 atoms   | FEF, $f = 0.14$                          | $1.75 \pm 0.31$ | $9.15 \pm 0.43$                      | 90.15  | 176                  | 0/8  |
|                         | FEF, $f = 0.25$                          | $1.75 \pm 0.25$ | $9.06 \pm 0.44$                      | 212.79 | 287                  | 0/8  |
|                         | logistic $T^* = 0.5$ eV                  | $2.12 \pm 0.30$ | $9.09 \pm 0.39$                      | 53.18  | 347                  | 0/8  |
|                         | logistic $T^* = 5.0$ eV                  | $2.62 \pm 0.50$ | $8.64 \pm 0.24$                      | 2.30   | 48                   | 0/8  |
|                         | ground state only ( $\lambda \equiv 0$ ) | $2.12 \pm 0.30$ | $8.57 \pm 0.54$                      | 0.00   | 0                    | 0/8  |
|                         | EAM (Marinica)                           | $3.12 \pm 0.30$ | $9.81 \pm 0.59$                      | 0.00   | 0                    | 0/8  |
| 2 keV<br>21 296 atoms   | FEF, $f = 0.14$                          | $2.62 \pm 0.32$ | $18.27 \pm 0.61$                     | 210.57 | 342                  | 0/8  |
|                         | logistic $T^* = 0.5$ eV                  | $3.00 \pm 0.19$ | $17.93 \pm 0.76$                     | 152.47 | 647                  | 0/8  |
|                         | logistic $T^* = 5.0$ eV                  | $3.50 \pm 0.42$ | $18.09 \pm 0.46$                     | 4.83   | 81                   | 0/8  |
|                         | ground state only ( $\lambda \equiv 0$ ) | $4.50 \pm 0.78$ | $17.96 \pm 0.94$                     | 0.00   | 0                    | 0/8  |
|                         | EAM (Marinica)                           | $5.62 \pm 0.37$ | $19.72 \pm 0.68$                     | 0.00   | 0                    | 0/8  |
| 5 keV<br>43 904 atoms   | FEF, $f = 0.14$                          | $5.25 \pm 0.62$ | $43.62 \pm 1.20$                     | 597.35 | 802                  | 0/12 |
|                         | logistic $T^* = 0.5$ eV                  | $5.20 \pm 0.71$ | $43.01 \pm 1.25$                     | 537.62 | 1550                 | 2/12 |
|                         | ground state only ( $\lambda \equiv 0$ ) | $8.23 \pm 0.63$ | $45.79 \pm 1.27$                     | 0.00   | 0                    | 0/13 |
|                         | EAM (Marinica)                           | $9.25 \pm 0.73$ | $52.83 \pm 2.59$                     | 0.00   | 0                    | 0/8  |

on the anneal-integrated convention they do not — both sit above it — so on that convention our switch is the more aggressive of the two and we do not claim otherwise. What does not depend on the convention is how the exposure is delivered: at 0.5 keV the peak excitation is 98.1 atoms for the free-energy-fraction switch against 193.2 for the logistic — fewer atoms carried further, or more atoms carried less, which is the distinction §5 turns on.

*The declared criterion never fires. A second failure mode does, twice, and only for the logistic — but the difference is not significant, and adding repetitions lowered the rate rather than raising it.* The pre-registered criterion is the original’s own: a series terminates when more than a quarter of its repetitions show an increasing thermal-spike volume after adaptive time stepping. Across the entire campaign — 181 cascades spanning every method and every energy — 0 repetitions are flagged by it. On that criterion alone the stability comparison is unresolved.

It is not the only way these dynamics fail. At 5 keV, 2 of the 12 repetitions of the reproduced logistic at  $T^* = 0.5$  eV diverged. In the worse of the two the maximum local atomic temperature reached  $1.9 \times 10^8$  K against  $1.2 \times 10^6$  K for its ground-state twin — the same starting configuration and the same recoil direction — the cell’s total energy moved by  $50.056$  eV atom<sup>−1</sup>, and it recorded 219 surviving Frenkel pairs against 9. The declared criterion catches neither, because the spike volume of a diverged run *decreases*; the divergence happens during the anneal rather than as a growing spike.

We therefore added a second detector, after seeing the first such run and disclosed as such: a cascade has failed if the maximum excursion of the cell’s total energy exceeds  $1$  eV atom<sup>−1</sup>. The

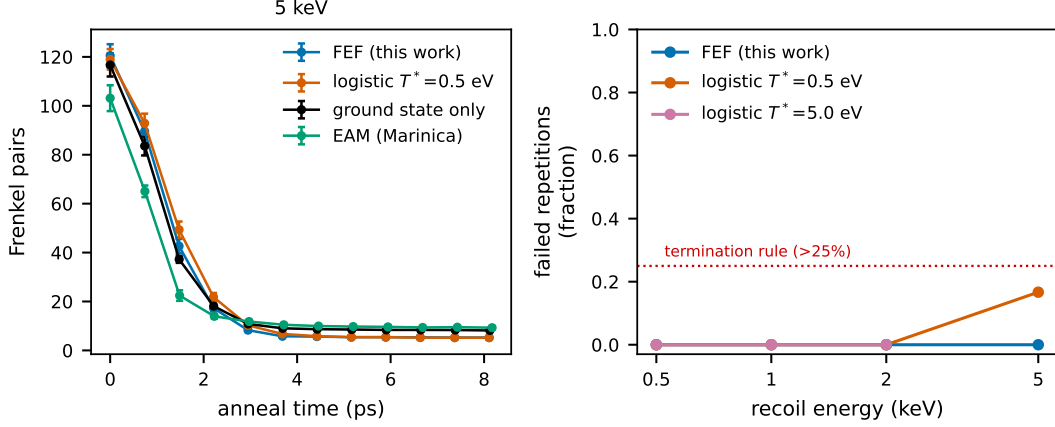


Figure 7: **Left:** surviving Frenkel pairs as a function of anneal time at the top of the ladder, which shows how far recombination has proceeded when the count is read off. **Right:** the fraction of repetitions that fail, under either the declared spike-volume criterion or the total-energy-divergence detector of §4.5; the termination rule thresholds this at one quarter. Diverged repetitions are excluded from the left panel.

threshold is not tuned. Across the campaign that excursion is a tight population between  $0.081$  and  $0.122 \text{ eV atom}^{-1}$  for every converged run of every method at every energy, while the two diverged runs sit at  $1.95$  and  $50.06 \text{ eV atom}^{-1}$  —  $16\times$  and  $410\times$  the largest converged excursion. The two populations are separated by more than an order of magnitude but not by an arbitrary margin: any threshold from about  $0.2$  to  $1.9 \text{ eV atom}^{-1}$  gives the same answer, and it is the milder of the two divergences, not the spectacular one, that sets the upper end.

*What this does and does not establish.* Every failure in the campaign — all 2 of them — belongs to the reproduced logistic at the top of the ladder; the free-energy-fraction switch, on the identical configurations and directions, shows a maximum excursion of  $0.122 \text{ eV atom}^{-1}$  there, indistinguishable from ground-state-only dynamics — and it does so while carrying greater exposure on the anneal-integrated convention over the converged runs, though less on the adaptive-stepping convention ( $131.78$  against  $233.74$ ) and less than the logistic’s all-repetition figure. The exposure comparison at this energy depends on both choices and we do not rest anything on it. But 2 failures in 12 is a rate of 17 %, and the published rule requires *more* than a quarter, so even the combined criterion does not terminate the series. And  $2/12$  against  $0/12$  is not a significant difference: Fisher’s exact test gives  $p = 0.48$ . We extended this energy from eight repetitions to 12 specifically to try to settle it, and the extension moved the estimate the wrong way for us: the four added logistic repetitions all converged, so the rate fell from a quarter to 17 % and the published criterion is now further from firing than it was. The honest summary is that the reproduced logistic exhibits a failure mode at 5 keV that our switch does not exhibit on the same configurations, that this is what the mechanism of §5 predicts, and that 12 repetitions at one energy are still not enough to call it a measured difference in stable recoil range. Reaching that would need either more repetitions or the higher recoil energies where the original reports its own series terminating, and neither was affordable here.

What does not require the instability to be triggered at all is §5: the free-energy-fraction switch spreads  $\lambda 1.9$  times less widely across the sample and shows 1.9 times smaller total-energy excursion than the logistic in unthermostatted dynamics, both with no overlap between the repetitions of the two switches. The net irreducible non-conservative work falls by 2.8 times as well, but at 3 repetitions that ratio is not resolved and we do not lean on it.

#### 4.6 The physical effect, in part

The signature of the layered framework is that sampling the high-temperature surface reduces both the surviving damage and the size of the thermal spike. That signature has to survive, or a switching function that improves stability has simply turned the physics off. Figure 8 is the test, and the third panel is the one that matters: the difference in Frenkel-pair count against ground-state-only dynamics, repetition by repetition, so the paired variance cancels.

*The Frenkel-pair half of the signature is reproduced; the spike-volume half is not resolvable either way.* Pooling the recoil energies by inverse variance — each energy contributing its paired difference as a fraction of the ground-state mean at that energy, which is the properly powered version of the per-energy comparison — the free-energy-fraction switch reduces the surviving Frenkel-pair count by  $34 \pm 9\%$  ( $z = -3.94$ , over 0.5, 1, 2, 5 keV). The reproduced logistic at  $T^* = 0.5$  eV reduces it by  $28 \pm 8\%$  ( $z = -3.54$ ) over the same four energies, and at  $T^* = 5.0$  eV by  $13 \pm 14\%$  ( $z = -0.91$ ) over three. Both switching functions therefore reproduce the framework’s signature reduction in surviving defects, and we say so: this axis is a check that our construction does not turn the physics off, and it passes for the baseline as well.

*The two are not distinguishable from each other.* Comparing them pooled would be the wrong test, since both are measured against the same ground-state runs. Compared directly, repetition by repetition and configuration by configuration, the free-energy-fraction switch leaves  $-0.12 \pm 0.30$  Frenkel pairs relative to the logistic over 34 paired cascades — negative meaning fewer defects for ours — with  $z = -0.40$  and a Wilcoxon  $p = 0.51$ . The paired difference favours the calibrated switch at 0.5, 1, 2 keV ( $-0.25, -0.38, -0.38$  pairs) and the logistic at 5 keV ( $+0.40$ ), so it does not even consistently point one way; on damage the two are not separable. The only axis on which they separate unambiguously is non-conservation, where the excursion and the spatial spread of  $\lambda$  differ with no overlap between repetitions (§5); the accuracy axis depends on which reference is believed (§4.3) and the divergence difference is not significant (§4.5). For scale against the reference rather than against the logistic, at 5 keV the free-energy-fraction switch gives  $5.25 \pm 0.62$  pairs against  $8.23 \pm 0.63$  for ground-state-only dynamics on the same configurations.

At 5 keV the logistic’s statistics are taken over the 10 of 12 repetitions that converged, the other 2 having diverged (§4.5); a diverged run’s defect count is meaningless and including it would inflate the logistic’s mean rather than deflate it.

The peak thermal-spike volume shows nothing resolvable. Pooled, the free-energy-fraction switch gives  $-2 \pm 3\%$  ( $z = -0.81$ ) and the logistic  $-3 \pm 3\%$  ( $z = -1.07$ ). Both point slightly in the direction the original reports, and neither is distinguishable from zero at this sample size; we cannot confirm the reduction in spike volume, and we cannot exclude one of the size reported either.

*The spike-volume measure is not robust to the probe radius, which weakens both our null and the original’s positive result.* The alpha-shape volume depends on a probe radius  $R_{\text{probe}}$ , taken as 1 nm above and in the original. Repeating the identical paired construction at 5 Å and 15 Å, the pooled change against ground-state-only dynamics is  $+5 \pm 2\%$  ( $z = +2.92$ , over 3 energies) and  $-8 \pm 4\%$  ( $z = -1.87$ , over 2) for our switch, and  $+6 \pm 2\%$  ( $z = +3.09$ ) and  $-12 \pm 5\%$  ( $z = -2.51$ ) for the logistic. *The sign of the effect follows the probe radius rather than the switching function, and it does so significantly in both directions:* at 5 Å both switches significantly *increase* the measured spike volume, at the nominal 1 nm neither changes it, and at 15 Å the logistic significantly *decreases* it ( $z = -2.51$ ) with our switch just short of significance ( $z = -1.87$ ). A quantity that returns a significant increase and a significant decrease on the same trajectories depending on an analysis parameter cannot adjudicate a physical claim in either direction, and we draw none from it. The Frenkel-pair count, which carries no such parameter, is what the physical-effect result rests on.

That the measurement can resolve a difference of the relevant size is established by the classical comparison, which is resolved in the opposite direction: the Marinica EAM potential produces  $+23 \pm 8\%$  more Frenkel pairs ( $z = +2.77$ ) and  $+18 \pm 4\%$  more spike volume ( $z = +4.51$ ) than the ground-state ACE potential on the identical cascades. The Frenkel-pair counts are settled well before the reported anneal time: over the last 2 ps of the anneal no group’s mean moves by more than 8.9% of its final value, so the 8 ps anneal is not truncating recombination at these energies.

The excited-state exposure rules out the trivial explanation. A switch that bought its stability by collapsing towards  $\lambda \rightarrow 0$  would show  $\chi \rightarrow 0$ ; ours is matched to the most aggressive published logistic by construction, and on the anneal-integrated convention it carries *more* exposure than that logistic at every energy of the ladder — by a factor of 1.8 at the foot and 1.1 at the top, the ratio narrowing as the recoil energy rises (Fig. 9) — with a peak excitation of 176.4 atoms at 1 keV against 347.3 for that logistic. The ratio is not constant, so “matched” means matched at one energy on one convention rather than uniformly; what holds uniformly is that it never drops below one, which is what rules out the collapse.

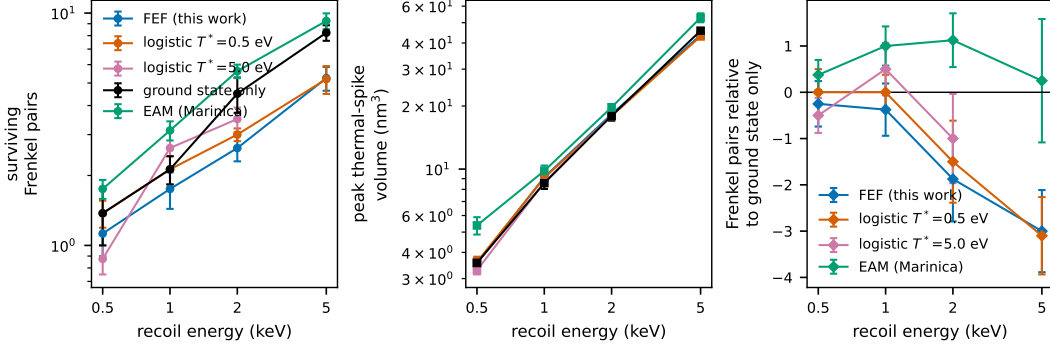


Figure 8: Cascade observables against recoil energy. **Left:** surviving Frenkel pairs. **Centre:** peak thermal-spike volume. **Right:** the Frenkel-pair count relative to ground-state-only dynamics, paired repetition by repetition; negative is the framework’s signature. Error bars are standard errors over the converged repetitions at each energy (Table 2).

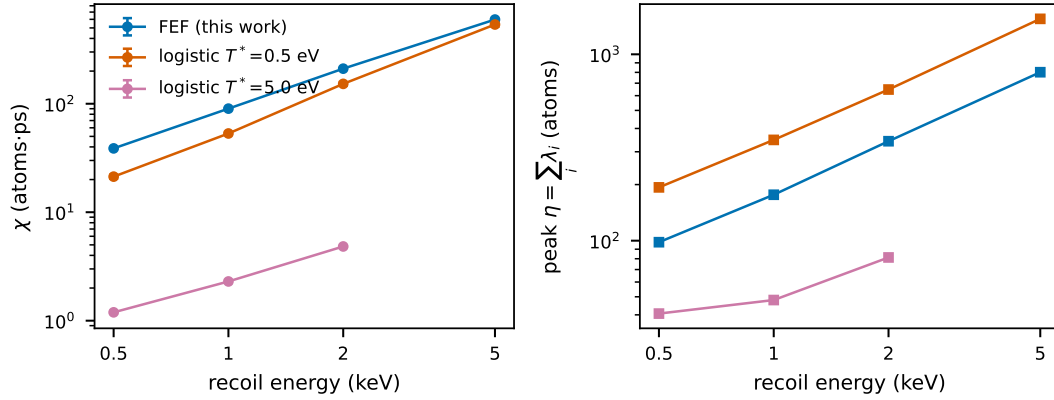


Figure 9: Excited-state exposure. **Left:**  $\chi = \int \sum_i \lambda_i dt$  against recoil energy. **Right:** the peak instantaneous excitation  $\eta = \sum_i \lambda_i$ . The partition slope was chosen to match the logistic’s exposure at the lowest energy, but on the shorter anneal of the calibration scan (§4.5); on the production anneal plotted here our switch runs above the logistic at every energy, by  $1.8\times$  at the foot of the ladder falling to  $1.1\times$  at the top. What the right panel shows is that it gets there differently: fewer atoms carried further rather than more atoms carried less.

## 5 Mechanism: where the instability comes from

§2.2 claims that the entire departure of the layered scheme from ground-state dynamics is  $d(K + U^{\text{gs}})/dt = \sum_j \lambda_j \Delta \mathbf{F}_j \cdot \mathbf{v}_j$ , and that the irreducible part of it is proportional to the spatial spread of  $\lambda$ . This section measures both.

**Protocol.** The original study’s own energy-conservation test: a 2000-atom cell, a 1 keV recoil, adaptive time stepping, pure microcanonical dynamics with no thermostat and no other external constraint, so every change in the total energy belongs to the scheme. Positions, velocities and the per-atom blend fraction are dumped along the trajectory; the per-atom energies *and forces* under each endpoint potential are then recovered by LAMMPS rerun, which gives both sides of the identity independently and without any modelling step. Three repetitions.

**The prediction that a uniform blend is conservative, tested exactly.** Equation (5) says that if  $\lambda$  is the same for every atom the scheme has no non-conservation at all, and that statement can be checked *without any time integral*:  $K + U^{\text{gs}} + \lambda \Delta U$  should be constant frame by frame. Running the two uniform corners of the family —  $\lambda \equiv 0$  and  $\lambda \equiv 1$  — and reconstructing that quantity from the recovered per-atom energies gives a drift of  $2.7 \times 10^{-4}$  eV atom<sup>-1</sup> and  $1.2 \times 10^{-4}$  eV atom<sup>-1</sup> over the whole trajectory, which is the adaptive integrator’s own noise. This simultaneously validates the

reconstruction (per-atom energies, kinetic-energy conversion, frame alignment) and confirms the prediction at integrator precision.

**With a spatially varying blend, the identity accounts for the drift.** The measured change of  $K + U^{\text{gs}}$  tracks the accumulated  $\int \sum_j \lambda_j \Delta \mathbf{F}_j \cdot \mathbf{v}_j dt$  to 0.471 of the peak excursion for the logistic and 0.613 for the free-energy-fraction switch (Fig. 10, centre). That residual is quadrature error in the right-hand side, not a failure of the identity — the uniform-corner test above has no quadrature and no residual — and it shrinks with the sampling rate: on a dedicated trajectory analysed at 3 dump intervals it falls from 4.97 at 40 steps to 0.13 at 4 steps for the logistic, and from 10.18 to 0.12 for the free-energy-fraction switch — more than an order of magnitude for a ten-fold refinement. At the finest sampling the two sides of the identity agree in their accumulated total to 60.02 eV against 56.89 eV for the logistic. The integrand is sampled on a trajectory whose time step spans six orders of magnitude, so it converges slowly; the left-hand side is exact at every frame.

**The magnitude, and the split.** Under the logistic at  $T^* = 0.5$  eV the maximum excursion is  $0.1047 \text{ eV atom}^{-1}$ , about  $5 \times$  the  $0.02 \text{ eV atom}^{-1}$  the original reports for the same test at the same  $T^*$  — the closest thing to a reproduction check available for this quantity, and not a close one. Our cell is smaller and our recoil lower, both of which raise the per-atom excursion, so we read the comparison as order-of-magnitude agreement rather than reproduction. Under the free-energy-fraction switch it is  $0.0546 \text{ eV atom}^{-1}$ , smaller by a factor of 1.9.

Equation (5) splits the injected work into a part that a uniform blend fraction would also produce — and that a Hamiltonian formulation would absorb — and a part that exists only because  $\lambda$  differs from atom to atom. Measured over the trajectory, the logistic injects 1.59 eV through the first channel and 68.41 eV through the second, with  $\langle \text{sd}_j \lambda_j \rangle = 0.0719$ ; the free-energy-fraction switch injects 0.60 eV and 24.63 eV with  $\langle \text{sd}_j \lambda_j \rangle = 0.0385$ . The net injection through the irreducible channel is therefore smaller by a factor of 2.8, tracking the 1.9-fold reduction in the spread of  $\lambda$  that Eq. (5) says controls it. *That factor is the least secure number in this section and we give it with its spread:* over 3 repetitions the logistic injects  $68.4 \pm 8.2$  eV and our switch  $24.6 \pm 19.6$  eV, and one of the three free-energy-fraction repetitions comes out at  $-13.6$  eV — the opposite sign. The reduction is therefore not resolved at this sample size, and a reader should weight it well below the two ratios that are: the total-energy excursion and the spatial spread of  $\lambda$  are separated with no overlap between the repetitions of the two switches. The *rectified* churn through that channel,  $\int |P_{\text{fluc}}| dt$ , is similar for the two (1059.74 against 985.00 eV): both switches move energy back and forth vigorously, and the difference is in how much fails to cancel.

**Why, and not at matched exposure.** The partition slope was set by matching  $\chi$  in the cascade protocol, but this test is a different cell at a different recoil energy and the match does not carry over: here the logistic delivers 103.5 and the free-energy-fraction switch 190.6 atoms.ps, so ours carries  $1.84 \times$  the exposure. That runs against us and it is worth being explicit that the comparison is therefore *a fortiori* rather than matched: our switch injects less energy while sampling the excited surface more, which is a stronger statement than the matched-exposure prediction of §2.2 and not a test of it. They also deliver that exposure differently. The logistic drives 465.7 atoms past  $\lambda = 0.9$  and accumulates a total blend-fraction variation  $\sum_i \int |d\lambda_i| = 3304$ ; the free-energy-fraction switch saturates 139.3 atoms, spreads the same exposure over 2000 atoms at a mean peak blend fraction of 0.206, and accumulates 3407. *The last of those numbers goes the wrong way for us and we flag it:* §2.2(iii) argues that a non-saturating switch drives each atom over a smaller *range* of  $\lambda$ , which should show up as a smaller traversal, and the measured traversal is slightly *larger* than the logistic’s rather than smaller. The hysteresis-loop picture is therefore not what distinguishes these two switches. What does is where in the sample the variation sits: the same total traversal spread thinly across many atoms costs less than the same traversal concentrated in a few, because (5) charges the spatial spread of  $\lambda$  and not its time-integrated motion. The same picture appears in the declared partition-slope scan, where at matched  $\chi$  the logistic reaches a peak  $\eta = \sum_i \lambda_i$  of 196 against 97 for the free-energy-fraction switch.  $\eta$  is a sum of blend fractions rather than a count of atoms, so the gloss that carries the mechanism is the pair above it: 2000 atoms at a mean peak blend fraction of 0.206 against 465.7 driven past  $\lambda = 0.9$  — against roughly twice the atoms carried half as far.

The mechanism is therefore not simply “long residency on  $\mathcal{U}_{\text{ht}}$ ”. Residency is what  $\chi$  measures, and at matched  $\chi$  the switch with the broader, gentler profile injects less energy, because the injection counts the *contrast* in  $\lambda$  across the sample and the *traversal* of  $\lambda$  in time, not the time spent at it.

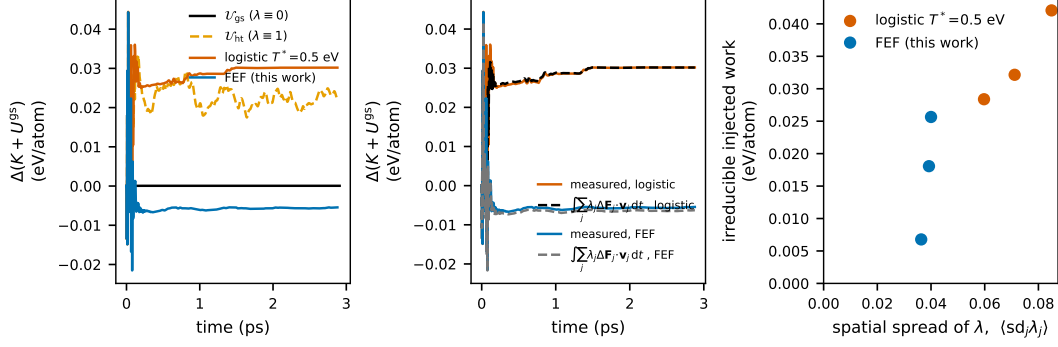


Figure 10: Non-conservation of the force-blended layered potential, measured in unthermostatted microcanonical dynamics. **Left:** excursion of  $K + U^{\text{gs}}$  for the two endpoint potentials ( $\lambda \equiv 0$  and  $\lambda \equiv 1$ , both exactly conservative in the sense of §2.2(i)) and for the two switches. **Centre:** the measured excursion against the right-hand side of Eq. (4), computed independently from per-atom forces recovered by `rerun`. **Right:** irreducible injected work against the spatial spread of  $\lambda$  — the quantity Eq. (5) says controls it — one point per repetition for each switch.

What destabilises the dynamics is driving atoms repeatedly across the full range of  $\lambda$  at a sharp front, which is exactly what a switch whose entire unit slope sits in a window of width  $T^*/3.5$  does to every atom the spike boundary sweeps past.

## 6 What did not work

### 6.1 The declared validation gate failed in the force channel

Before computing anything we committed to a gate on the tight-binding reference: it would be used only if its distortion at  $\theta_{\text{ht}}$  correlated with the ACE inter-potential distortion at Pearson  $r \geq 0.7$  in the force channel, with an implied endpoint blend fraction in  $[0.7, 1.4]$ . The measured values are  $r = 0.182$  and  $\lambda^*(\theta_{\text{ht}}) = 0.136$  in the force channel: the gate fails, and by a wide margin. In the energy channel the endpoint blend fraction is 0.993 — within 1 % of the value of unity that it must take if the surrogate is describing the same physical distortion as the two potentials — but the correlation, 0.605, still misses the threshold we set.

We did what the pre-registration says to do when the gate fails: the deployed calibration is the citation-anchored one, fixed by the reported intermediate-temperature optimum rather than by our own measurement, and the measured curve is reported beside it as an independent estimate that disagrees. We resisted the temptation to relax the gate after seeing the numbers. One consequence is worth stating plainly: because the deployed calibration is anchored to a number we cannot re-derive, the accuracy result in §4.3 is weaker evidence than it would have been with the reference sets in hand.

### 6.2 Two independent estimates of the calibration disagree by a factor of four

Our surrogate says the correct blend fraction at 0.1 eV is 0.0547 (force channel) to 0.0673 (energy channel), and a second, direction-independent estimator built from distortion magnitudes rather than projections agrees (0.0866 and 0.0496). The reported optima are 0.35 and 0.20. The disagreement is not a detail: it is the difference between a quadratic and a linear dependence on electronic temperature, and the quadratic one is what the Sommerfeld expansion requires of any metal at electronic temperatures small compared with the scale of structure in the density of states.

We can see three explanations and cannot decide between them. (i) *The surrogate is wrong in a way the energy-channel endpoint check does not catch.* A  $d$ -band model with canonical ratios has no  $s$ - $d$  hybridisation and no self-consistency; if real tungsten has fine density-of-states structure within 0.1 eV of  $E_{\text{F}}$  that our 80 meV-broadened model washes out, the true distortion would grow faster than  $\theta^2$  and the optimum would rise. We tested the sensitivity to both calibration targets and to the  $k$ -mesh (§4.2); the curve moves, but not by a factor of four. (ii) *The reported optimum is not measuring the electronic free-energy distortion.* The two provided potentials are independent

fits with different radial bases (dispersion parameters 0.7075 and 0.5128) and different selection criteria, so their difference contains a fitting component that is not electronic-temperature physics. A least-squares blend fraction against an intermediate reference will absorb part of that component. Our force-channel measurement is consistent with this reading: only 3% of the variance of the ACE force difference is explained by the distortion we compute. (iii) *Both*. Deciding requires the reference sets.

### 6.3 A driver map we rejected on paper, and one we rejected in the code

The pre-registration records two candidate driver maps set aside before any run. The energy-equipartition map  $\theta_e = \sqrt{\theta_{gs}^2 + \kappa \langle T_i - T_0 \rangle_+}$ , obtained by equating the local electronic Sommerfeld energy  $\frac{1}{2} \gamma T_e^2$  to a fraction of the local ionic excess, fails on arithmetic that needs no simulation: because the electronic heat capacity is small, the square root rises with near-infinite slope just above onset, so any  $\kappa$  large enough to reach  $\lambda \sim 1$  inside a thermal spike also gives  $\lambda \sim 0.3$  for an ordinary 1600 K equilibrium fluctuation. The alternative  $T_e = T_i$  — no partition slope at all — is arguably the physically correct statement, and it is what our calibration curve would be evaluated at in true thermal equilibrium; but it assigns  $\lambda = 0.29$  to a cell sitting at the 1000 K bath, which denies that  $\mathcal{U}_{gs}$  is the right ambient potential and breaks comparability with the framework and with the baseline. We report it as a remark and do not use it.

A third thing failed in implementation rather than in principle: LAMMPS atom-style variables do not accept `min()` or `max()`, so the first realisation of the switch ran zero steps. The guards are now written with boolean sub-expressions, and `code/cascade/verify_switch.py` checks the LAMMPS field against the Python reference implementation atom by atom on a configuration whose driver spans 1000 K to  $3.54 \times 10^5$  K; the largest discrepancy is  $5.0 \times 10^{-13}$ .

### 6.4 What we could not reach

The recoil ladder stops at 5 keV. The instability the original work documents appears at 15 keV for its most aggressive setting, in cells two orders of magnitude larger than ours. §4.5 reports what the termination criterion says over the ladder we could afford; where no series terminates, we say that the stability comparison is unresolved at these energies rather than claiming a result, and fall back on the mechanism measurements, which are direct and do not require the instability to be triggered.

Finally, our excited-state exposure  $\chi$  is not numerically comparable to the original’s. We define it explicitly as  $\int \sum_i \lambda_i dt$  in atoms-ps, integrated by trapezoid on a trace sampled every ten steps, and we use it only for matched-aggressiveness comparisons among our own runs. We note in passing that the original’s own  $\chi$  values and its  $\eta(t)$  traces do not appear mutually consistent when read off the published figures, which is one reason we did not attempt to match its absolute scale.

## 7 Relation to existing work

**Electron temperature in radiation-damage molecular dynamics.** The standard route is the two-temperature model Kaganov et al. [1957], Anisimov et al. [1974], coupled to molecular dynamics by Duffy and Rutherford [2007] and Rutherford and Duffy [2007] and applied to tungsten cascades by Zarkadoula et al. [2015], in which an electronic continuum exchanges energy with the ions through a friction term and a coupling constant  $G$ , and fast ions lose energy to electronic stopping Caro and Victoria [1989], Ziegler et al. [2010]. What such schemes move between the subsystems is *energy*; the interatomic potential is never modified, so the ions always move on the ground-state Born–Oppenheimer surface, and effects that change the *shape* of that surface — bond softening or hardening at elevated electronic temperature Recoules et al. [2006] — are outside their reach. The layered construction Wood et al. [2026] addresses precisely that gap, and the present work is about the rule that decides how far along the distortion each atom is placed. The two are complementary rather than competing: nothing here prevents a two-temperature energy exchange being run alongside, and §8 argues that it should be.

**Potentials that depend on electronic temperature.** The temperature-dependent potentials of Ackland [2012] add an explicit  $T_e^2 N(E_F)$  term with a parameterised pairwise density of states; Khakshouri et al. [2008] built, for tungsten specifically, an analytic potential whose parameters are

continuous functions of  $T_e$  fitted to finite-electronic-temperature density-functional theory. That work is the closest ancestor of the layered construction and, to our reading, is not cited by it. The difference is where the approximation sits: a single  $T_e$ -parameterised analytic form must be accurate at every electronic temperature simultaneously, whereas a two-endpoint blend confines the approximation entirely to the interpolation weight and lets each endpoint be as accurate as a machine-learned model allows. That is the trade this paper takes as given; our contribution is that the weight can then be *derived* rather than assumed.

On the machine-learned side, Srinivasan et al. [2024] fit moment tensor potentials at Chebyshev nodes in  $T_e$  and interpolate with a Chebyshev polynomial, and Zhang et al. [2020] train deep-potential models on finite-temperature density-functional data. Both interpolate numerically, with weights that are basis coefficients carrying no physical meaning, and both need more than two potentials. Our claim is narrower and testable: with the blend fraction set to the realised fraction of the electronic free-energy distortion, two endpoints suffice over this range. Ben Mahmoud et al. [2022] show that hot-electron free energies can be predicted from ground-state data at arbitrary  $T_e$ , which raises the fair question of why one should layer at all; the answer here is cost and cascade-scale tractability, and it is an answer that should be argued rather than assumed.

**Blending two potentials, and the force it does not conserve.** Interpolating between two models over a spatial or per-atom variable is the adaptive-resolution problem. Force interpolation (AdResS Praprotnik et al. [2005]) is not conservative and needs a thermodynamic force to keep the density uniform; energy interpolation (H-AdResS Potestio et al. [2013]) restores a Hamiltonian at the price of a drift term of exactly the form our identity takes in the slowly-varying limit,  $\sum_i \Delta U_i \dot{\lambda}_i$ . The same dichotomy appears in adaptive QM/MM partitioning Buló et al. [2009], Heyden et al. [2007], Kerdcharoen and Morokuma [2002] and, for interatomic potentials specifically, in the adaptive-precision framework of Immel et al. [2025a], whose more recent variant Immel et al. [2025b] folds the coupling into a single Hamiltonian and conserves energy by construction. The machinery there looks identical to the layered potential and the *meaning* is not: in adaptive precision the two endpoints describe the same physics and their disagreement is error, so the switch may be tuned for efficiency; here they describe two different electronic states and their disagreement is signal, so the blend fraction has to be physically right. That is why the present work spends its effort on calibrating  $\lambda$  rather than on conserving energy — and why we regard the Hamiltonian route as the natural next step rather than a competitor (§8).

**Concurrent work.** Kümmel and Roth [2026] run two-temperature molecular dynamics for copper with a  $T_e$ -dependent potential and an energy-conservation algorithm. Their emphasis is the integrator and the pressure relaxation; ours is the choice of blend fraction and its validation at an intermediate electronic temperature. The materials also differ in a way that matters: copper is free-electron-like with a filled  $d$  band, whereas tungsten’s  $d$ -band pseudogap makes both its electronic heat capacity and its electron–phonon coupling strongly non-free-electron Lin et al. [2008] — which is exactly why the calibration curve of §4.2 departs from the naive linear interpolation in  $T_e$ .

## 8 Limitations

**The first-principles reference sets were not available.** The accuracy axis of this study was designed around three author-provided VASP reference sets at  $T_e = 0.001, 0.1$  and  $0.5$  eV, of which the intermediate one is the genuinely held-out temperature. Those files were not available to us (§3.2), and no electronic-structure engine was either. Everything we report on the accuracy axis therefore rests either on a tight-binding surrogate that we built for the purpose or on the two blend-fraction optima quoted from the original supplemental material. Neither substitutes for the measurement that was intended. The consequences are specific and we have tried not to blur them: the surrogate validates cleanly in the energy channel and fails its own declared gate in the force channel; the quoted optima cannot be re-derived, re-checked, or extended to other electronic temperatures. If the reference sets become available, the single most valuable follow-up is to recompute Fig. 3 against them — the code path is `code/derivation/calibrate_lambda.py` with the reference distortion swapped in, and nothing else changes.

**The surrogate is a  $d$ -band model, not density-functional theory.** It is calibrated to two measured properties of tungsten and it reproduces the endpoint blend fraction to within 1 % in the energy

channel without being told to, which is a non-trivial check. It nonetheless resolves the *direction* of the per-atom force distortion poorly (§4.2), and we cannot separate how much of that is the model’s crudeness from how much is a genuine mismatch between the two fitted potentials. Both readings are stated; only one of them is a criticism of the framework, and we do not assert which.

**Cells and recoil energies are far smaller than the original’s.** The original study used cells from  $2.5 \times 10^5$  to  $1.6 \times 10^7$  atoms and recoil energies to 200 keV on leadership-class hardware. We had one 32-vCPU node. Absolute defect counts are not comparable to the original and we do not present them as such; every claim we make is a comparison between methods run in the same cell, from the same starting configuration, with the same recoil direction. The stability question in particular lives at the top of the recoil spectrum, which we cannot reach: §4.5 reports what the criterion says over the ladder we could afford and does not extrapolate past it.

**The partition slope is a model, not a measurement.** The driver map contains one scalar,  $f$ , and the two-temperature estimate of it (§4.4) spans more than an order of magnitude, mostly because the electron–phonon coupling of tungsten is itself uncertain by a factor of four between the value used throughout the cascade literature and the modern ab-initio and pump–probe values Fujimoto et al. [1984], Daraszewicz et al. [2014], Lin et al. [2008]. We therefore fix  $f$  by matching the excited-state exposure  $\chi$  to the reproduced logistic, which is what the head-to-head comparison requires anyway, and report the physical bracket alongside rather than pretending the two agree. A switching function keyed to a locally estimated electronic temperature inherits whatever error is in that estimate; this is a real limitation of the whole approach, not of our particular form.

**Half of the reported physical signature is not reproduced, and the stability evidence is an existence proof rather than a termination energy.** We recover the reduction in surviving Frenkel pairs; we cannot resolve the reduction in peak thermal-spike volume, for any switching function at any energy we could reach — and we show that this measure cannot decide the question either way, returning a significant increase at one alpha-shape probe radius and a significant decrease at another on the same trajectories (§4.6). Nor does the original’s own termination criterion fire anywhere, so we have no measured stable-recoil-range endpoint. What we have is 2 diverged repetitions out of 12 for the reproduced logistic at the top of our ladder — a rate of 17 %, below the published rule’s quarter — against none for our switch on the identical configurations, a difference that Fisher’s exact test puts at  $p = 0.48$ . We extended that energy from eight repetitions to 12 to try to resolve it and the estimate moved away from us, not towards us. That is an existence proof of a failure mode, not a termination energy and not a significant difference in rate, and we do not dress it up as either. Both of these could be consequences of the cell sizes and recoil energies we could afford — the effects the original reports grow steeply with recoil energy — but that is a conjecture, and the measurements we have say what they say.

**The blend is still not conservative.** We quantify the non-conservation exactly and we reduce it, but  $\sum_j \lambda_j \Delta \mathbf{F}_j \cdot \mathbf{v}_j$  does not vanish while  $\lambda$  varies in space, and in a cascade it must. The clean fix is known: fold the coupling into a Hamiltonian, as the conservative adaptive-precision scheme does Immel et al. [2025b], at the cost of a  $\nabla \lambda$  term in the forces and of the generality that the hybrid/scaled mechanism buys. Our result should be read as saying that the switching function is worth getting right *whether or not* one takes that step: a Hamiltonian formulation removes the drift but does not tell you what blend fraction to use, and an uncalibrated blend fraction is wrong even when it is conservative.

## 9 Conclusion

The auxiliary variable of a two-state layered potential is a physical quantity with a measurable value, and treating it as one removes both free parameters of the published switching function. Fixing the blend fraction to the realised fraction of the electronic free-energy distortion makes the transition width a consequence of the Sommerfeld law rather than a choice, separates the calibration question from the electron–ion coupling question, and — through the exact non-conservation identity  $d(K + U^{\text{gs}})/dt = \sum_j \lambda_j \Delta \mathbf{F}_j \cdot \mathbf{v}_j$ , whose irreducible part is set by the spatial spread of  $\lambda$  — explains why a calibrated switch is also a gentler one. In a dedicated unthermostatted test the resulting switch halves the total-energy excursion of the scheme and spreads  $\lambda$   $1.9\times$  less widely across the sample at

greater excited-state exposure; across 181 cascades of tungsten up to 5 keV it reduces the surviving Frenkel-pair count by  $34 \pm 9\%$  relative to ground-state-only dynamics — all without a switching temperature. The reproduced logistic reduces it by  $28 \pm 8\%$ , and on damage the two are not separable at this sample size. Where they separate unambiguously is non-conservation, with no overlap between the repetitions of the two switches on either the energy excursion or the spatial spread of  $\lambda$ . The accuracy axis separates them too, but in a direction that depends on which reference is believed, and a failure mode appeared only for the logistic — 2 times in 12 repetitions, which is an existence proof and not a significant difference in rate.

Three things we found cut against the framework rather than for it, and they matter more than the improvement. First, the electronic free energy of tungsten is quadratic in the electronic temperature over the range in question, so the correct blend fraction at 0.1 eV is of order 0.0547–0.0673, not the 0.2 that naive linear-in- $T_e$  reasoning suggests, and not the 0.2–0.35 spanned by the two reported optima. Second, only about 3% of the variance of the *force* difference between the two provided potentials is explained by the electronic-temperature distortion our surrogate computes, against 37% of the energy difference. Since the framework blends forces, most of what the blend does to the dynamics may be the difference between two independently optimised fits rather than electronic-temperature physics. Third, the reduction in peak thermal-spike volume that the framework reports is not resolved in our cascades for any switching function at any energy we could reach — and the measure itself is not trustworthy in either direction, returning a significant increase at one alpha-shape probe radius and a significant decrease at another on the identical trajectories, so what we really report is that this quantity cannot settle the question. All three are testable claims, all three are testable with data that already exists, and all three should be tested before the layered construction is used quantitatively.

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## A Reproducing every number

The pre-registration (proposal/code/method\_plan.md) was committed before any production run; its commit and the recorded timestamp are in proposal/logs/method\_plan\_timestamp.txt. Every quantitative statement in this paper is a  $\LaTeX$  macro generated by proposal/code/analysis/make\_numbers.py from a logged record under proposal/logs/ — a JSON result file for everything measured in this study, or one of the raw run logs kept under proposal/logs/provenance/ for the few setup quantities (the relaxed lattice constants, the node throughput, the evaluator speed-up) that come from the equilibration and benchmark runs. Every measured quantity — every entry in every table and figure, and every result quoted in the text — is generated, and none is typed by hand. Protocol constants that define the runs rather than report them (bath temperatures, ensemble compositions, grid ranges, the recoil ladder), values attributed to the original work, and the two model scalars of the surrogate are written directly in the source; code/analysis/verify\_paper.py lists every numeral typed into the prose so that this boundary can be inspected rather than trusted. The pipeline is

1. `code/derivation/make_ensemble.py` — the evaluation ensemble (94 configurations) by  $\mathcal{U}_{\text{gs}}$  molecular dynamics;
2. `code/derivation/eval_ace.py` — both potentials evaluated on it by LAMMPS rerun, giving the inter-potential distortion;
3. `code/derivation/tb_finite_te.py` and `calibrate_lambda.py` — the finite-electronic-temperature reference and the blend-fraction sweep ( $\rightarrow$  `logs/calibration_main.json`);
4. `code/analysis/switch_compare.py` — the head-to-head accuracy comparison ( $\rightarrow$  `logs/switch_accuracy.json`);
5. `code/cascade/run_campaign.py` — cell construction and the cascade grid ( $\rightarrow$  `logs/cascades/*.json`, one record per run, and `logs/fscan/`);
6. `code/analysis/collect.py` — aggregation ( $\rightarrow$  `logs/cascade_summary.json`);
7. `code/analysis/mechanism.py` and `wellbehaved.py` — the non-conservation and well-behavedness measurements;
8. `code/analysis/figures.py` and `make_numbers.py` — figures and the macro file.

`code/cascade/verify_switch.py` checks that the LAMMPS realisation of each switching function equals its Python reference implementation atom by atom on a real configuration, and reports the analytic well-behavedness checks. `proposal/attempts_log.md` records what was tried and abandoned, including the two driver maps rejected before any run and the changes made after the pre-registration was committed.